

THE WEIHRAUCH DEGREES OF PROBLEMS OF CATEGORICITY

NIKOLAY BAZHENOV, JOSIAH JACOBSEN-GROCOTT, LILING KO, HEER TERN KOH, AND KENG MENG NG

Abstract. In this article, we analyse the uniform computational content of categoricity problems, showing relationships between the complexity of Scott families and their Weihrauch degrees. We give upper bounds for some classes of linear orders and Boolean algebras, and show in some cases that these are sharp. We also explore the question of which Weihrauch degrees are realizable as categoricity problems, and show that no Weihrauch degree between LLPO and WKL are realizable. In this article, we also explore the poset of Weihrauch degrees of categoricity problems of computable structures, showing that it is closed with respect to product and parallelization, and that there exists a greatest degree in this poset. In contrast, we show that there is no single structure whose categoricity problem bounds all other categoricity problems. This motivates the introduction of the categoricity problems of classes of structures, which we study in relation to other notions of universality in the literature.

§1. Introduction. In computable structure theory [4, 20, 35, 36], one topic that has been widely studied is: “How hard it is to find isomorphisms between two given isomorphic structures?”

The motivation for studying this topic goes back to the following old observation: in general, computable isomorphic copies of the same structure $\mathcal{A}$ can have *different* algorithmic properties. This kind of behavior was already witnessed by Fröhlich and Shepherdson [25], and by Mal’tsev [34]. We give a simple folklore example of a ‘badly behaved’ $\mathcal{A}$. One can build a computable isomorphic copy $\mathcal{B} = (\mathbb{N}, \leq_B)$ of the linear order $\mathcal{A} = (\mathbb{N}, \leq)$ such that its adjacency relation

$$\text{Adj}(\mathcal{B}) = \{(a, b) : (a <_B b) \& \neg \exists c (a <_B c <_B b)\}$$

is not computable. On the other hand, the order $(\mathbb{N}, \leq)$ itself has computable adjacency relation.

Mal’tsev [34] isolated a class of structures $\mathcal{A}$ such that all computable isomorphic copies of $\mathcal{A}$ share *the same* computability-theoretic properties. He called those structures *autostable*, but modern sources typically call them *computably categorical*. Over the years, this approach has led to the definition of several interesting properties of structures.

DEFINITION 1.1. Let α be a non-zero computable ordinal. A computable structure $\mathcal{M}$ is:

- *computably categorical* (Δ_α^0 -categorical) if for any pair of computable copies of $\mathcal{M}$, there is a computable (Δ_α^0) isomorphism between those copies.
- *uniformly computably categorical* if there a Turing functional, Φ , such that for any pair of computable copies, (X, Y) of $\mathcal{M}$ we have that $\Phi^{(X, Y)}$ is an isomorphism from X to Y .

For a computable structure $\mathcal{M}$ we say that the *degree of categoricity* (if it exists) is the least Turing degree $\mathbf{d}$ such that for all pairs of computable copies of $\mathcal{M}$, there is a $\mathbf{d}$ -computable isomorphism between those copies [21].

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These notions have been extensively studied, and the study of what structures have a degree of categoricity, and what Turing degrees can be realized as the degree of categoricity of a structure have occupied a fundamental part of the study of computable structure theory—we refer to, e.g., [21, 28, 1, 13, 22, 12, 14, 15]. For example, it is known that for every Π_1^0 function singleton f satisfying $f \geq_T \mathbf{0}''$, the degree $\deg_T(f)$ is a degree of categoricity [15]. For a detailed exposition of results in this area, we refer to the surveys [23, 6].

One can think of the notion of being Δ_α^0 -categorical as a relativization of the notion of being computably categorical, while keeping the copies of $\mathcal{M}$ computable. If we instead consider copies of $\mathcal{M}$ of arbitrary complexity, we get the following notion.

DEFINITION 1.2. A computable structure $\mathcal{M}$ is:

- *relatively computably categorical* (*relatively Δ_α^0 -categorical*) if for any pair of copies of $\mathcal{M}$, (X, Y) , there is an (X, Y) -computable $(\Delta_\alpha^0(X, Y))$ isomorphism between X and Y .
- *relatively uniformly computably categorical* if there a Turing functional, Φ , such that for any pair of copies, (X, Y) of $\mathcal{M}$ we have that $\Phi^{(X, Y)}$ is an isomorphism from X to Y .

It turns out that being relatively uniformly computably categorical and uniformly computably categorical are equivalent notions (this follows, e.g, from [41], see Corollary 2.6 in [18]), but when the uniformity is dropped, these notions can differ. From the definition, it is not hard to see that if $\mathcal{M}$ is relatively Δ_α^0 -categorical, then $\mathcal{M}$ is Δ_α^0 -categorical. However, for every $\alpha < \omega_1^{CK}$ there is a computably categorical structure that is not relatively Δ_α^0 -categorical [19].

While we can define the degree of categoricity of a structure, relative to some oracle X , this degree is always going to be above the Turing degree of X , so there is not going to be a Turing degree that can compute isomorphism for all oracles. Work by Csimá and Harrison-Trainor [16] overcomes this issue by showing that on a cone, the degree of categoricity relative to an oracle X stabilizes as $X^{(\alpha)}$ for some $\alpha < \omega_1$.

In this paper, we appeal to the Weihrauch degrees as an alternative way to relativize the degree of categoricity for a structure. The Weihrauch degrees have been used to classify problems in computable analysis and provide a perspective on which problems are harder. For a fixed structure $\mathcal{M}$ there is a natural problem, and thus Weihrauch degree, that measures how hard it is to find isomorphisms between arbitrary (and not just computable) copies of $\mathcal{M}$.

We refer to [8] for the background on Weihrauch degrees. To discuss our results, we first give the following definitions.

DEFINITION 1.3. A function $f : \subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ is *Weihrauch reducible* to $g : \subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$, written $f \leq_W g$ if there exist computable functionals Φ, Ψ such that for any $x \in \text{dom}(f)$, $\Phi(x) \in \text{dom}(g)$, and for any $y \in g\Phi(x)$, $\Psi(y, x) \in f(x)$. Here $x, \Phi(x)$ are generally referred to as *instances* of f, g respectively, and $\Psi(y, x), y$ are generally referred to as *solutions* of $x, \Phi(x)$ respectively.

Intuitively, if $f \leq_W g$, then for any instance x of f , we are able to produce a solution to x provided that we can do the same for g . In this way, f is easier to ‘solve’ than g . Under this setting, we study the following problem.

DEFINITION 1.4. Given a countable model $\mathcal{M}$ with (at most) countable language $\{R_0, \dots\}$ we define the problem $\text{cat}(\mathcal{M})$ as follows:

- $p \oplus q \in \mathbb{N}^\mathbb{N}$ is an *instance* of $\text{cat}(\mathcal{M})$ if $\mathcal{M} \cong (\mathbb{N}; R_0^p, \dots) \cong (\mathbb{N}; R_0^q, \dots)$ (where $R_e^p(\bar{a}) \iff p(\langle e, \bar{a} \rangle) = 1$, and $R_e^q(\bar{a}) \iff (q(\langle e, \bar{a} \rangle) = 1)$).
- $f \in \mathbb{N}^\mathbb{N}$ is a *solution* to $\text{cat}(\mathcal{M})$ if and only if f is an isomorphism from $(\mathbb{N}; R_0^p, \dots)$ to $(\mathbb{N}; R_0^q, \dots)$.

In proofs we will often refer to p and q as if they are the models $(\mathbb{N}; R_0^p, \dots)$ and $(\mathbb{N}; R_0^q, \dots)$.

Unlike the case with the degree of categoricity being measured by the Turing degrees, for all structures $\mathcal{M}$, $\text{cat}(\mathcal{M})$ has a well-defined Weihrauch degree. Following the research trends in computable structure theory, we also study which Weihrauch degrees are degrees of $\text{cat}(\mathcal{M})$ for some $\mathcal{M}$, and show that there are many Weihrauch degrees that are not.

In Section 2 we look at relatively computable categorical structures and consider what the Weihrauch degree of $\text{cat}(\mathcal{M})$ could be in this case. We are able to show that uniformly computably categorical models are precisely those for whom $\text{cat}(\mathcal{M}) \equiv_W \text{id}$, and that for a structure, $\mathcal{M}$, being relatively computable categorical is equivalent to having $\text{cat}(\mathcal{M}) \leq_W \mathbf{C}'_{\mathbb{N}}$.

In Section 4 we consider WKL, the problem of finding paths in infinite binary trees, and ask if WKL can be equivalent to $\text{cat}(\mathcal{M})$ for some $\mathcal{M}$. As we show, there is in fact no structure $\mathcal{M}$ such that $\text{LLPO} \leq_W \text{cat}(\mathcal{M}) \leq_W \text{WKL}$, giving some examples of Weihrauch degrees that are not realizable as $\text{cat}(\mathcal{M})$ for some $\mathcal{M}$.

In Section 3 we compute the Weihrauch degree of $\text{cat}(\omega)$, $\text{cat}(\zeta)$, $\text{cat}(\omega + \omega)$ among other linear orders, giving examples of structures where $\text{cat}(\mathcal{M})$ is equivalent to $\mathbf{C}_{\mathbb{N}}$, lim , and $\mathbf{C}_{\mathbb{N}} \star \text{lim}$.

We then go on in Section 5 to generalize these ideas and give upper bounds for the degrees of $\text{cat}(\alpha)$ for $\alpha < \omega_1^{CK}$ and $\text{cat}(\mathcal{S})$ for computable scattered linear orders $\mathcal{S}$.

In Section 6 we use the Scott complexity approach to extend the results of Section 5 to arbitrary relatively Δ^0_α -categorical structures. This allows us to compute the Weihrauch degrees of $\text{cat}(\beta)$ for all computable ordinals $\beta < \omega^\omega$. We also find the Weihrauch degrees of $\text{cat}(\mathcal{B})$ for some familiar Boolean algebras $\mathcal{B}$.

In Section 7 we consider the structure $\mathbf{CAT}_0$ which contains the Weihrauch degrees of $\text{cat}(\mathcal{M})$ for all computable structures $\mathcal{M}$. In particular, we show that $\mathbf{CAT}_0$ is closed with respect to the product of problems $\times$. We also prove that the poset of c.e. Turing degrees is embeddable into $\mathbf{CAT}_0$. In section 10 we prove that $\mathbf{CAT}_0$ has a maximal element, while also showing that there is maximum Weihrauch degree among the degrees of $\text{cat}(\mathcal{M})$ for all structures $\mathcal{M}$.

Given there is no upper bound among the degrees of $\text{cat}(\mathcal{M})$, a natural question one may ask, is if there is a general Weihrauch degree that acts as an upper bound on $\text{cat}(\mathcal{M})$? The answer is yes, as illustrated by the general categoricity problem, cat :

DEFINITION 1.5. We define the problem $\text{cat} := \bigcup_{\mathcal{M}} \text{cat}(\mathcal{M})$.

If $\mathcal{M} \not\cong \mathcal{N}$, then $\text{dom}(\text{cat}(\mathcal{M})) \cap \text{dom}(\text{cat}(\mathcal{N})) = \emptyset$, so the identity function serves to show that $\text{cat}(\mathcal{M}) \leq_W \text{cat}$ for any structure $\mathcal{M}$. In Section 8 we look at this problem and prove that it is equivalent to $\mathbf{C}_{\mathbb{N}^{\mathbb{N}}}$, the problem of finding paths in ill-founded trees on $\mathbb{N}^{<\mathbb{N}}$. The proof of this fact involves constructing a special tree, $\mathcal{U}$, that we call the universal tree. This tree has the unique property that if T is computable and ill-founded, then $T \times \mathcal{U} \cong \mathcal{U}$.

In Section 9, we look at what happens to the general problem when we restrict to a class of structures $\mathbb{K}$, by defining $\text{cat}(\mathbb{K}) := \bigcup_{\mathcal{M} \in \mathbb{K}} \text{cat}(\mathcal{M})$. This gives us notion of universal class:

DEFINITION 1.6. We say that a class $\mathbb{K}$ is *on top for cat* if $\text{cat} \leq_W \text{cat}(\mathbb{K})$.

It turns out this is a distinct notion from other definitions of universality, such as being on top for effective bi-interpretability or on top for Borel reducibility, but it is unclear how exactly on top for cat relates to these notions. The natural universal classes all seem to be on top for cat.

While the general problem cat may not have been studied before, problems that involve finding isomorphisms have shown up in the literature. One such example is CWO:

DEFINITION 1.7. The problem **CWO** is the problem whose instances are pairs of well orders X, Y and whose solutions are an embedding of X as an initial segment of Y or an embedding of Y as an initial segment of X .

It should be clear that $\text{cat}(\text{WO}) = \bigcup_{\alpha < \omega_1} \text{cat}(\alpha) \leq_W \text{CWO}$ via the identity reduction. **CWO** has been studied in [32] and [27]. Its Weihrauch degree contains many interesting problems such as **ATR**, the problem version of the reverse mathematics principle, and $\text{UC}_{\mathbb{N}\mathbb{N}}$, the problem of finding infinite paths in trees with exactly one infinite path. If one looks at the proof (Lemma 5.7 in [32]), that reduces these problems to **CWO**, one can observe that the forward functional only produces well orders of equal length, and thus we have that $\text{cat}(\text{WO})$ is equivalent to **CWO**.

§2. The degrees of computably categorical models.

THEOREM 2.1. *If $\mathcal{M}$ is uniformly computably categorical, then $\text{cat}(\mathcal{M}) \equiv_W \text{id}$.*

PROOF. $\mathcal{M}$ has a computable instance, so $\text{cat}(\mathcal{M}) \geq_W \text{id}$. For the other direction observe that there is a partial computable functional Φ_e such that if p, q are computable instances of $\mathcal{M}$ then $\Phi_e(p \oplus q)$ produces an isomorphism between these models. But the computable copies of $\mathcal{M}$ form a dense subset of $\text{dom}(\text{cat}(\mathcal{M}))$ so Φ_e can be turned into a functional that works on all of $\text{dom}(\text{cat}(\mathcal{M}))$. $\dashv$

THEOREM 2.2. *For a computable model $\mathcal{M}$ the following are equivalent:*

1. $\mathcal{M}$ is relatively computably categorical.
2. $\text{cat}(\mathcal{M}) \leq_W \liminf_{\mathbb{N}}$.
3. $\text{cat}(\mathcal{M}) \leq_W P$ for some first order problem P (first order means that the solutions to P are essentially members of $\mathbb{N}$ rather than $\mathbb{N}^{\mathbb{N}}$).

PROOF. (3) $\iff$ (1) follows from the definitions. (2) $\implies$ (3) as $\liminf_{\mathbb{N}}$ is a first order problem.

We show that (1) $\implies$ (2). Suppose that $\mathcal{M}$ is relatively computably categorical and let $\{\varphi_e\}_{e \in \mathbb{N}}$ be a list of all partial computable functions. Given a potential instance $p \oplus q$, all we need to do is to find an index e so that $\varphi_e^{p \oplus q}$ is an isomorphism from p to q , provided $p \oplus q \in \text{dom}(\text{cat}(\mathcal{M}))$. Define f as follows. For each e , let $n_e[s]$ be the least such that for each $n < n_e[s]$, $n \in \text{dom}(\varphi_{e,s}^{p \oplus q})$ and $n \in \text{range}(\varphi_{e,s}^{p \oplus q})$. Let $f(0) = 0$, and for each $s > 0$, define $f(s)$ to be the least e such that $\varphi_{e,s}^{p \oplus q}$ currently looks to be a partial isomorphism, and such that $n_e[s] > n_e[s-1]$. If $p \oplus q \in \text{dom}(\text{cat}(\mathcal{M}))$, then it follows that $\liminf_s f(s)$ exists. Furthermore, it produces the least index e for which $\varphi_e^{p \oplus q}$ is an isomorphism from p to q as desired. $\dashv$

THEOREM 2.3. *There is a model $\mathcal{M}$ such that $\text{cat}(\mathcal{M}) \equiv_W \mathcal{C}'_{\mathbb{N}}$.*

PROOF. The definition of $\mathcal{M}$ is as follows. We use a language with two unary functions which we will call 0 and 1. There will be a fixed generator called $\emptyset$ that generates all the other members of our $\mathcal{M}$ from the two unary functions. In this way we can think of each $\sigma \in 2^{<\mathbb{N}}$ as corresponding to an element of our model. This mapping is onto, but there will be some $\sigma \neq \tau$ such that $(\emptyset)\sigma = (\emptyset)\tau$ (we are applying our function from the right so that $(\emptyset)(\sigma \cap \tau) = ((\emptyset)\sigma)\tau$). In fact, our model will be completely determined by what σ and τ become equivalent.

To make the construction a little clearer, we say $(\emptyset)\sigma \cong_n \emptyset$ if for all $\tau, \rho \in 2^{<n}$ we have $(\emptyset)\sigma\rho = (\emptyset)\sigma\tau$ if and only if $(\emptyset)\rho = (\emptyset)\tau$. We use $\cong$ in place of $\cong_{\omega}$. Essentially $(\emptyset)\sigma \cong \emptyset$ means there is a monomorphism (of $\mathcal{M}$) taking $\emptyset$ to $(\emptyset)\sigma$.

$\mathcal{M}$ is defined to have the minimum set of equivalences satisfying the following (see Fig. 1):

- $(\emptyset)0 \cong \emptyset$.

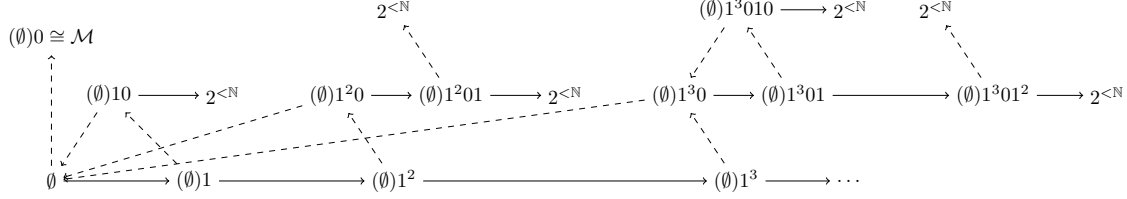


FIGURE 1. $\mathcal{M}$ such that $\text{cat}(\mathcal{M}) \equiv_{\mathbf{W}} \mathbf{C}'_{\mathbb{N}}$. We denote the functions 0 and 1 with dashed and solid arrows respectively. $2^{<\mathbb{N}}$ represents the full binary tree.

- $(\emptyset)1^n 0 \cong_n \emptyset$ for all $n > 0$
- $(\emptyset)1^n 00 = \emptyset$ for all $n > 0$

Observe that $\mathcal{M}$ is rigid and that any isomorphism is determined by where we send $\emptyset$. $\emptyset$ is not the only generator of $\mathcal{M}$; since $\emptyset = (\emptyset)1^n 00$, $(\emptyset)1^n 0$ and $(\emptyset)1^n$ are also generators, but they have different atomic types compared to $\emptyset$. In the type space of $\mathcal{M}$ we have that $\text{type}(\emptyset) = \lim_{n \in \mathbb{N}} \text{type}((\emptyset)1^n 0)$. As illustrated in Fig. 1, the structure ‘rooted’ at $(\emptyset)1^n 0$ resembles $\mathcal{M}$ up to a depth of n .

Now for the reduction. Let p be the standard computable copy of $\mathcal{M}$. The idea with this reduction is that we build a copy q of $\mathcal{M}$ where the root of q codes a solution to $\mathbf{C}'_{\mathbb{N}}$. The key idea here is that we can change our mind about where the root is in 2 different ways. The first way is we can add a new root and have it point to the old root via 0. The other way is to remove a choice of root, but saying that it is actually $(\emptyset)1^n 0$ for some large enough n .

Let $r = \oplus_{n \in \mathbb{N}} r_n$ be an instance of $\mathbf{C}'_{\mathbb{N}}$. We define q in stages. At each stage we have some choice of root, $\emptyset_s$. If the root does not change between stages then we continue adding elements as if we are building the standard copy.

We will also need to maintain a guess of solution to r . Let m_s be least m such that $r_s(m) \neq r_{s+1}(m)$. Then we let $c_s = \min(\mathbb{N} \setminus \text{range}(r_s \upharpoonright m_s))$. So $\liminf_{s \in \mathbb{N}} c_s$ will be a solution to r .

The construction goes as follows. At stage $s + 1$ we ask to compare c_{s+1} and c_s and do one of three things:

- If $c_s = c_{s+1}$, then we have $\emptyset_{s+1} = \emptyset_s$ and we continue building from $\emptyset_s$.
- If $c_s < c_{s+1}$, then we pick a new number for $\emptyset_{s+1}$ and define $\emptyset_s = (\emptyset_{s+1})0$. We then associate c_{s+1} with $\emptyset_{s+1}$.
- If $c_s > c_{s+1}$, then we remove the old root by choosing n large enough that $(\emptyset_s)01^n 0$ has not been defined and set it equal to $\emptyset_s$. This ensures that $\emptyset_s$ cannot be the true root.

We can repeat this process until we are left with a root that is associated with some number $\leq c_{s+1}$. If this root is associated with c_{s+1} , then it will be $\emptyset_{s+1}$. If not, then we will add a new root to be $\emptyset_{s+1}$ and associate it with c_{s+1} .

For the backwards functional we simply look at where $\emptyset^p$ is sent and follow the construction of q to see what number c is associated with it. c will be a solution to r . $\dashv$

Our examples of $\text{cat}(\mathcal{M})$ in this section have all landed at the level of id , $\mathbf{C}_{\mathbb{N}}$, or $\mathbf{C}'_{\mathbb{N}}$. We make the following conjecture

CONJECTURE 2.4. If $\mathcal{M}$ is a relatively computably categorical structure, then $\text{cat}(\mathcal{M}) \in \{\text{id}, \mathbf{C}_{\mathbb{N}}, \mathbf{C}'_{\mathbb{N}}\}$.

§3. Linear order examples. In this section, we look at instances of $\text{cat}(\mathcal{M})$ where $\mathcal{M}$ is some specific linear order and try to work out where they belong in the Weihrauch degrees. We will prove all the results given in Figure 2.

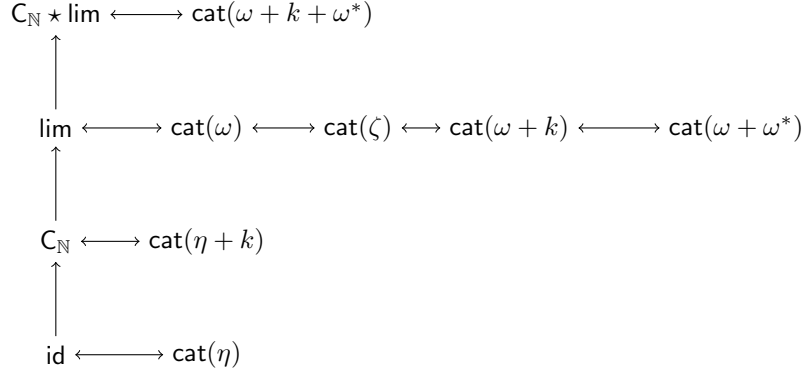


FIGURE 2. Weihrauch degrees of categoricity for various linear orders.

Since η is uniformly computably categorical, Theorem 2.1 tells us that $\text{id} \equiv_W \text{cat}(\eta)$. For the other computably categorical linear orders, Theorem 2.2 tells us that these are all below $\mathbb{C}'_{\mathbb{N}}$, but they turn out to be equivalent to $\mathbb{C}_{\mathbb{N}}$ without the need for a jump.

THEOREM 3.1. *If $\mathcal{M}$ is a computably categorical linear order and $\mathcal{M} \not\equiv \eta$, then $\text{cat}(\mathcal{M}) \equiv_W \mathbb{C}_{\mathbb{N}}$.*

PROOF. To start with, it is known from [30, 39] that the computably categorical linear orders all have the form $k_0 + \eta + \dots + \eta + k_n$ where $k_0, \dots, k_n$ are finite (possibly empty) linear orders. We will give reductions just for the order $\eta + 2 + \eta$, but the other reductions can be built using the same idea.

Proof of $\text{cat}(\eta + 2 + \eta) \leq_W \mathbb{C}_{\mathbb{N}}$. The forward functional will be given an instance $p \oplus q$ and needs to produce an instance, $r \in \mathbb{N}^{\mathbb{N}}$ of $\mathbb{C}_{\mathbb{N}}$ whose solution will identify where the 2-block is within p and q respectively. To do this, we encode each selection of $a, b \in p$ and $c, d \in q$ as a single natural number.

We build r in stages and place $\langle a, b, c, d \rangle$ into its range if we witness that one of our possible selections (a, b, c, d) is *invalid*. A selection (a, b, c, d) is invalid if one of the following is observed:

- $a \not\leq_p b$ or $c \not\leq_q d$.
- $\exists e[a <_p e <_p b]$
- $\exists e[c <_q e <_q d]$

This is a c.e. condition to check so we can add an invalid selection to the range of r as it is observed to be invalid. If a selection is not invalid, then it must be that b is the p -successor of a and d is the q -successor of c . So a solution to r must select the 2-blocks as desired.

For the backwards functional, we are given p, q and a valid selection (a, b, c, d) and must produce an isomorphism from p to q . It is clear we must have $a \mapsto c$ and $b \mapsto d$. Observe that $\{n : n <_p a\}, \{n : n <_q c\}$ are both computable in $p \oplus q$ and has order type η . The standard back and forth construction suffices to produce an isomorphism between these sets. An isomorphism between $\{n : n >_p a\}, \{n : n >_q c\}$ can be defined in a similar way.

Proof of $\mathbb{C}_{\mathbb{N}} \leq_{sW} \text{cat}(\eta + 2 + \eta)$. For the forward functional, from an instance r of $\mathbb{C}_{\mathbb{N}}$, we produce two linear orders p, q such that an isomorphism between them encodes a number not in the range of r . To do this, we will make p a standard copy where the 2-block is given by a known pair a, b .

When building q we will start off by picking a pair $a_0 = \langle 0, 0 \rangle, b_0 = \langle 0, 1 \rangle$ and build q as if b_0 is the successor of a_0 . If at some stage we observe $0 \in \text{range}(r)$ then we start adding elements in between a_0 and b_0 and pick a new pair $a_1 = \langle 1, i \rangle, b_1 = \langle 1, i + 1 \rangle$. We can choose i to be large enough that a_1, b_1 have not yet been placed in q . We then proceed to treat b_1 as the successor of a_1 and build copies of η on either side of this pair.

If at some later stage we see $1 \in \text{range}(r)$ then we repeat the process and chose new witnesses a_2, b_2 . Since $\text{range}(r) \neq \omega$ this process will eventually stop and there will be some n such that b_n remains the successor of a_n at all later stages. An isomorphism from p to q must map a to $a_n = \langle n, i \rangle$ for some i . The backward functional need only look at the first coordinate of a_n to know that $n \notin \text{range}(r)$ and so can produce n as a solution to r . $\dashv$

Now that we have categorized the degrees of computably categorical linear orders we will start looking at discrete linear orders.

THEOREM 3.2. *The following problems are all Weihrauch equivalent:*

- lim .
- $\text{cat}(\omega)$.
- $\text{cat}(\zeta)$.
- $\text{cat}(\omega + \omega^*)$.
- $\text{cat}(\omega + k)$ for $k \in \mathbb{N}$.

PROOF. It is easy to see that given copies p, q of ω , we may produce copies p', q' of $\omega + k$, (and $\omega + \omega^*$) so that any isomorphism between p', q' may be restricted to produce an isomorphism between p, q . Thus, $\text{cat}(\omega) \leq_W \text{cat}(\omega + k), \text{cat}(\omega + \omega^*)$. We also have that $\text{cat}(\zeta) \leq_W \text{cat}(\omega) \times \text{cat}(\omega)$. The idea here is that given two copies of ζ we can select a center point on each of them, and look for two separate isomorphisms of ω using $\text{cat}(\omega) \times \text{cat}(\omega)$, one of the elements above the center and one of the elements below the center.

PROOF OF $\text{lim} \leq_{sW} \text{cat}(\zeta)$. For the forward functional we are given a sequence $(r_n)_{n \in \mathbb{N}} \subseteq \omega^\omega$ and need to build copies of ζ, p, q such that an isomorphism between them will be able to compute $r = \lim_{n \in \mathbb{N}} r_n$. We take p to be a standard copy, importantly one where $[[a, b]]$ is computable.

We build q by starting with $\dots a_{-1} < a_0 < a_1 < \dots$ a standard copy of ζ but will possibly add new elements between a_i and a_{i+1} for $i \in \mathbb{N}$. The idea is that we want to encode the settling time of $(r_n(i))_{n \in \mathbb{N}}$, so each time $r_n(i) \neq r_{n+1}(i)$ we add n elements between a_i and a_{i+1} in q . Since $(r_n)_{n \in \mathbb{N}}$ must be convergent, we only add finitely many elements between each a_i and a_{i+1} , and so, q is a copy of ζ . This describes the forward functional.

For the backward functional, we are given some isomorphism $f : p \rightarrow q$ and now need to compute r . Consider $s_i = [[f^{-1}(a_i), f^{-1}(a_{i+1})]]$. By our choice of p , s_i is computable uniformly in f . We claim that $r(i) = r_{s_i}(i)$. If not, then $(r_n(i))_{n \in \mathbb{N}}$ will change at some stage greater than s_i , and we would have added more than s_i many points in between a_i and a_{i+1} at that stage, a contradiction. $\dashv$

PROOF OF $\leq_W \text{lim}$. For this we will use some facts about lim : it is closed under parallelization, so there was no harm in using two copies of $\text{cat}(\omega)$ above, and lim is equivalent to $\text{jump} = \{(p, p') : p \in \mathbb{N}^\mathbb{N}\}$ where p' is the Turing jump of p . So for our purposes it is sufficient to show that $\omega, \omega + \omega^*, \omega + k$ are all uniformly computably categorical relative to a jump.

For a linear order $\mathcal{M}$, the successor relation $S(x, y) \iff [[x, y]] = 2$ is Π_1^0 and hence computable from a jump of $\mathcal{M}$. Similarly we have that the constants $\min(\mathcal{M})$ and $\max(\mathcal{M})$, if they exist, are computable from a jump of $\mathcal{M}$. When we add $\min, \max$ and S to the language, the linear orders

$\omega, \omega + \omega^*, \omega + k$ become finitely generated from $\min, \max$, and so they are uniformly computably categorical in this expanded language. $\dashv$

This completes the reductions. $\dashv$

THEOREM 3.3. *The following problems are all Weihrauch equivalent:*

- $\mathbb{C}_{\mathbb{N}} \star \lim$.
- $\text{cat}(\omega \cdot k)$ for $k > 1$.
- $\text{cat}(\omega + k + \omega^*)$ for $k > 0$.

PROOF. We break this proof into two parts. First:

PROOF OF $\leq_W \mathbb{C}_{\mathbb{N}} \star \lim$. Like in the proof of Theorem 3.2 we will use the fact that $\lim$ is equivalent to computing a jump and consider these linear orders with $\max, \min, S$ all being computable.

While these linear orders may be rigid, they are not finitely generated from $\min$ and $\max$ anymore, as we need to identify more points. In the case of $\omega + k + \omega^*$ we need to identify the starting point of the k -block and in the case of $\omega \cdot k$ we need to identify the start of all the ω blocks aside from the first block.

Each of these block is identified by a starting element that has no predecessor. Furthermore, all other elements, except for $\min$ will have a predecessor. So, like in the proof of Theorem 3.1, we can search for these points in a co-c.e. way, and thus reduce the problem to $\mathbb{C}_{\mathbb{N}}$. $\dashv$

Second we do:

PROOF OF $\mathbb{C}_{\mathbb{N}} \star \lim \leq_W \text{cat}(\omega + \omega)$. For the forward functional, we are given $(r_n)_{n \in \mathbb{N}}$, an instance of $\lim$, as well as a computable functional Φ . Our job is to build linear orders p, q such than an isomorphism with be able to compute not just $r = \lim_{n \in \mathbb{N}} r_n$ but also some $m \notin \text{range}(\Phi(r))$.

We have a tool to encode r from the proof of Theorem 3.2, so we need a way encode some $m \notin \text{range}(\Phi(r))$. Define u_n to be the least $i < n$ such that $r_n(i) \neq r_{n+1}(i)$ or n if no such i exists. Define $m_n = \min(\mathbb{N} \setminus \text{range}(\Phi_n(r_n \upharpoonright u_n)))$ (here Φ_n only runs n steps of computation and produces stops if it needs to use parts of the oracle beyond u_n). Since we have that $u_n \rightarrow \infty$ and $(r_n \upharpoonright u_n) \subset r$ infinitely often, we will have that $m = \liminf_{n \in \mathbb{N}} m_n$ exists and is not in $\text{range}(\Phi(r))$.

Now we get to the details of our construction. We will let p be a standard copy of $\omega + \omega$, i.e. one where the successor relation and the location limit points are computable. q with start of with, $a_0 < a_1 < \dots$, a standard copy of ω . This will be part of the upper ω -block and we will add points in between a_i and a_{i+1} to encode the settling time of $(r_n(i))_{n \in \mathbb{N}}$, like in the proof of Theorem 3.2.

We will build the lower ω -block in stages, only adding points to the end. To encode m we will use special witness elements for each natural number. At stage $n + 1$ of the construction we will have witness elements $b_{m_n} <_q \dots <_q b_0$. These will be the initial segment of upper ω -block at stage n . If $m_{n+1} > m_n$ Then we will extend this block by picking new witness elements. If $m_{n+1} < m_n$ then we will remove $b_{m_n}, \dots, b_{m_{n+1}+1}$ by making them part of the lower ω -block.

In this way, we will have ensured that the first limit point in p must be mapped to a copy of witness of m , so m is computable from an isomorphism. The same argument as in the proof of Theorem 3.2 will show how r is computable from an isomorphism. $\dashv$

The proof that $\mathbb{C}_{\mathbb{N}} \star \lim \leq_W \text{cat}(\omega + k + \omega^*)$ can be done in a similar way. The trick is that witness elements are now witness blocks, representing the entire k -block. Witness elements for smaller numbers than m_n will be considered a part of the lower ω -block and they will not have any new elements placed below them unless m_n decreases. Witness elements for numbers larger than m_n will be incorporated into the upper ω^* -block. To facilitate this, we will need to add the a_i over

time, rather than right away and this may increase the size of $[a_{i+1}, a_i]$, but that is not a problem, as we only need an upper bound on the settling time of $(r_n(i))_{n \in \mathbb{N}}$. $\dashv$

§4. Degrees below WKL.

THEOREM 4.1. *There exists a non-computably categorical structure $\mathcal{A}$ such that $\text{cat}(\mathcal{A}) \leq_W \text{WKL}$.*

PROOF. Consider the computable graph $\mathcal{A}$ defined as follows. The graph consists of countably many ‘components’, each of which shall be associated with the computation $\varphi_e(e)$, where φ_e is the e -th computable function. (See Figure 3.) Each component initially consists of three vertices

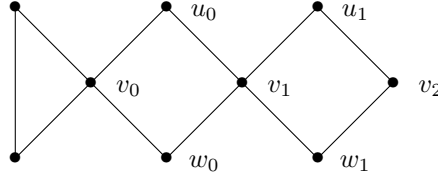


FIGURE 3. The first two components of the graph.

v_e, u_e, w_e . Once $\varphi_e(e) \downarrow$, we add a new vertex to the structure, connecting it to u_e . Evidently, $\mathcal{A}$ is a computable graph.

We first discuss why $\text{cat}(\mathcal{A}) \leq_W \text{WKL}$. Given any two presentations of $\mathcal{A}$, say B, C , observe that we are able to computably (in B or C) determine the components each vertex belongs to. To see this, note that v_0 is contained in the only triangle in the graph, and once v_0 is determined, we may uniquely determine each component of B and C . Without loss of generality, we shall fix u_e to always be of lower index than w_e in the respective presentations B, C . To avoid ambiguity, we denote each element with a superscript to indicate the presentation in which it belongs. For example, v_e^B denotes the unique element in the e^{th} component in B connected to vertices in the previous component, and u_e^B denotes the element in the e^{th} component in B of lower index that is adjacent to v_e^B .

Evidently, the challenge in computing an isomorphism between B and C is knowing whether the additional vertex, if $\varphi_e(e) \downarrow$, will be attached to u_e^B or w_e^B (in B) and u_e^C or w_e^C (in C). To decide, we shall utilise the given path h on the tree T (to be constructed by us). The idea is that if $h(e) = 0$, then we shall map u_e^B to u_e^C , and w_e^B to w_e^C . On the other hand, if $h(e) = 1$, then u_e^B will be mapped to w_e^C , and w_e^B will be mapped to u_e^C .

Under this setting, it is evident how T should be defined. At each stage s , we decide for each $\sigma \in 2^s$ whether $\sigma \in T$. For each e , wait for $\varphi_e(e) \downarrow$. If this never occurs, then no further action is required. If $\varphi_e(e) \downarrow$, then we further wait for the additional vertices to be added into the e^{th} component of B and C respectively. If $B, C \cong \mathcal{A}$, this must occur at some finite stage s . Once this is discovered then we perform one of the following actions.

- If the additional vertex is adjacent to u_e^B and u_e^C (or w_e^B and w_e^C) in B and C respectively, then for all $\sigma \in T_s$ of length s such that $\sigma(e) = 1$, define $\tau \notin T$ for all $\tau \supsetneq \sigma$.
- Otherwise, for all $\sigma \in T_s$ of length s such that $\sigma(e) = 0$, define $\tau \notin T$ for all $\tau \supsetneq \sigma$.

We thus have that T is computable in B and C . For any path h through T , consider the map as defined above: $u_e^B \mapsto u_e^C$ if $h(e) = 0$, and $u_e^B \mapsto w_e^C$ otherwise. If $\varphi_e(e)$ never converges, then

regardless of whether $h(e) = 0$ or $h(e) = 1$, the map is an isomorphism on the e^{th} component. Now consider the case where $\varphi_e(e) \downarrow$. If $h(e) = 0$, then by the definition of T , we must have witnessed that the additional vertex is adjacent to u_e^B and u_e^C , or w_e^B and w_e^C in B and C respectively. Thus, $u_e^B \mapsto u_e^C$ must also be an isomorphism on the e^{th} component. One may similarly verify that if $h(e) = 1$, then $u_e^B \mapsto w_e^C$ is also an isomorphism on the e^{th} component. Thus, we have that $\text{cat}(\mathcal{A}) \leq_W \text{WKL}$.

To complete the proof of the theorem, we now show that $\mathcal{A}$ is not computably categorical. In particular, we construct computable presentations $B, C \cong \mathcal{A}$ satisfying the following requirements.

R_e : $\varphi_e : B \rightarrow C$ is not an isomorphism.

To satisfy each requirement, we search for some *witness* x where either $\varphi_e(x) \uparrow$, or $\varphi_e(x) \downarrow$ but $\deg(x) \neq \deg(\varphi_e(x))$. We build B to be ‘standard’: the e^{th} component consists of 3 vertices given by the indices $4e, 4e+1, 4e+2$, where $4e$ represents the vertex u_e . The vertices with indices $4e+3$ will be used for the additional vertices to be added whenever we see some $\varphi_n(n) \downarrow$. Now consider the following function: $\varphi_{s(e,n)} = \varphi_e(4n)$. By the recursion theorem, for every e , there exists infinitely many n (computable in s) so that $\varphi_n(n) = \varphi_e(4n)$. In particular, we may pick for each e some $n_e > n_{e'}$ for all $e' < e$ such that $\varphi_{n_e}(n_e) = \varphi_e(4n_e)$. Obviously, this element $4n_e$ shall be the witness for R_e .

In C , we similarly define the e^{th} component to consist of the vertices with index $4e, 4e+1, 4e+2$, and use the vertices with indices of the form $4e+3$ to represent the additional vertices. As discussed, we use the n_e -th component to satisfy R_e . If $\varphi_{n_e}(n_e) \uparrow$, then no further action is required as we have that $\varphi_e(4n_e) \uparrow$. On the other hand, if $\varphi_{n_e}(n_e) \downarrow$, then we attach the additional vertex of the n_e -th component to $4e$ in B , but attach it to the vertex in the n_e -th component with an index different from $\varphi_{n_e}(n_e) \downarrow = \varphi_e(4n_e)$ in C . This ensures that φ_e cannot be an isomorphism as it maps a vertex with degree 3 in B to a vertex of degree 2 in C . $\dashv$

THEOREM 4.2. *There are no structures $\mathcal{A}$ for which $\text{LLPO} \leq_W \text{cat}(\mathcal{A}) \leq_W \text{WKL}$.*

PROOF. Suppose that there exists $\mathcal{A}$ such that $\text{LLPO} \leq_W \text{cat}(\mathcal{A}) \leq_W \text{WKL}$ as witnessed by Φ, Ψ and $\hat{\Phi}, \hat{\Psi}$ respectively. Let $\Phi(0^\mathbb{N}, 0^\mathbb{N}) = (B_0, B_1)$ where $B_0 \cong B_1 \cong \mathcal{A}$, and let $T = \hat{\Phi}\Phi(0^\mathbb{N}, 0^\mathbb{N})$ be an infinite binary tree. For each $s \in \mathbb{N}$, we use $B_0 \upharpoonright s$ and $B_1 \upharpoonright s$ to denote the first and second entries of $\Phi(0^s, 0^s)$ respectively. Let $s \in \mathbb{N}$ be such that both of the following holds:

- (1) For each $\sigma \in T \upharpoonright s = \hat{\Phi}\Phi(0^s, 0^s)$, either $\sigma \smallfrown 0, \sigma \smallfrown 1 \notin T \upharpoonright s$, or there exists a path in T extending σ .
- (2) For each $\sigma \in T \upharpoonright s$, let $f^\sigma = \hat{\Psi}(\sigma \oplus (B_0 \upharpoonright s, B_1 \upharpoonright s))$ be the partial isomorphism from B_0 to B_1 . If $\sigma \in T \upharpoonright s$ is extendable to a path in T , then $\Psi(f^\sigma \oplus (0^s, 0^s)) \downarrow$.

It is not hard to see why s satisfying (1) above exists. To see why s also satisfying (2) exists, first observe that if Φ, Ψ and $\hat{\Phi}, \hat{\Psi}$ are valid Weihrauch reductions, then for any path h through T , $\Psi(f^h \oplus (0^\mathbb{N}, 0^\mathbb{N})) \downarrow$. Next, by compactness and the finite use principle, we further have that there exists s , and $\sigma_0, \sigma_1, \dots, \sigma_n \in T \upharpoonright s$ such that for any path h in T , there exists some i such that $\Psi(f^h \oplus (0^\mathbb{N}, 0^\mathbb{N})) = \Psi(f^{\sigma_i} \oplus (0^s, 0^s)) \downarrow$. Thus, s satisfying both (1) and (2) must exist.

Now fix some $\sigma \in T \upharpoonright s$ that is extendable to a path in T , and assume without loss of generality that $\Psi(f^\sigma \oplus (0^s, 0^s)) \downarrow = 0$. We claim that f^σ is extendable to an isomorphism from C_0 to C_1 where $(C_0, C_1) = \Phi(0^t 1^\mathbb{N}, 0^\mathbb{N})$ for some $t \geq s$. (Note that C_0, C_1 extends $B_0 \upharpoonright \text{dom}(f_\sigma)$ and $B_1 \upharpoonright \text{range}(f_\sigma)$ respectively.) Once again, let $C_0 \upharpoonright t$ and $C_1 \upharpoonright t$ denote the first and second entries of $\Phi(0^t, 0^t)$ respectively. Fix some presentation $A \cong \mathcal{A}$ and search for $t \geq s$ such that all of the following holds:

- (3) Let $T_0 = \hat{\Phi}(B_0, A)$. For each $\tau \in T_0 \upharpoonright t = \hat{\Phi}(B_0 \upharpoonright t, A \upharpoonright t)$, either $\tau \smallfrown 0, \tau \smallfrown 1 \notin T_0 \upharpoonright t$, or τ extends to a path in T_0 .
- (4) For each $\tau \in T_0 \upharpoonright t$, let $g_0^\tau = \hat{\Psi}(\tau \oplus (B_0 \upharpoonright t, A \upharpoonright t))$ be the partial isomorphism from B_0 to A . If τ is extendable to a path in T_0 , then the domain of g_0^τ contains the domain of f^σ (which is finite).
- (5) Let $T_1 = \hat{\Phi}(B_1, A)$. For each $\gamma \in T_1 \upharpoonright t$, either $\gamma \smallfrown 0, \gamma \smallfrown 1 \notin T_1 \upharpoonright t$, or γ extends to a path in T_1 .
- (6) For each $\gamma \in T_1 \upharpoonright t$, let $g_1^\gamma = \hat{\Psi}(\gamma \oplus (B_1 \upharpoonright t, A \upharpoonright t))$ be the partial isomorphism from B_1 to A . If γ is extendable to a path in T_1 , then the domain of g_1^γ contains the range of f^σ . (Note that $\text{range}(f^\sigma)$ is finite.)

Since (B_0, A) and (B_1, A) are both valid instances of $\text{cat}(\mathcal{A})$, $\hat{\Phi}(B_0, A)$ and $\hat{\Phi}(B_1, A)$ must both be instances of WKL. Once again, the finite use principle and compactness of Cantor space guarantees that such a $t \geq s$ must exist.

Fix t satisfying properties (3)-(6) above. We claim that for such a t , there must exist $\tau \in T_0 \upharpoonright t$ that is extendable to a path in T_0 and $\gamma \in T_1 \upharpoonright t$ extendable to a path in T_1 which are also extendable to paths in $\hat{\Phi}(C_0, A)$ and $\hat{\Phi}(C_1, A)$ respectively. Suppose for a contradiction that every $\tau \in T_0 \upharpoonright t$ extendable to a path in T_0 is not extendable to a path in $\hat{\Phi}(C_0, A)$. Since C_0 extends $B_0 \upharpoonright t$ (recall that $B_0 \upharpoonright t$ is defined to be the first entry of $\Phi(0^t, 0^t)$, and C_0 is the first entry of $\Phi(0^t 1^\mathbb{N}, 0^\mathbb{N})$), the tree $\hat{\Phi}(C_0, A)$ must be an extension of $T_0 \upharpoonright t$. By (3), if $\tau \in T_0 \upharpoonright t$ is not extendable to a path in $T_0 = \hat{\Phi}(B_0, A)$, then both $\tau \smallfrown 0$ and $\tau \smallfrown 1$ are not in $T_0 \upharpoonright t$. In particular, every $\tau \in T_0 \upharpoonright t$ not extendable to a path in T_0 is also not extendable to a path in $\hat{\Phi}(C_0, A)$. Together with the assumption that every $\tau \in T_0 \upharpoonright t$ extendable to a path in T_0 is not extendable to a path in $\hat{\Phi}(C_0, A)$, we have that $\hat{\Phi}(C_0, A)$ has no paths, a contradiction to the assumption that $\hat{\Phi}$ produces an instance of WKL. A similar argument suffices to show that there is also some $\gamma \in T_1 \upharpoonright t$ that is extendable to a path in $\hat{\Phi}(C_1, A)$.

Now fix $\tau \in T_0 \upharpoonright t$ that is extendable to a path in both T_0 and $\hat{\Phi}(C_0, A)$. Let τ' and τ'' denote the respective paths. Also define $g_0 : B_0 \rightarrow A$ and $h_0 : C_0 \rightarrow A$ to be the isomorphisms given by $\hat{\Psi}(\tau' \oplus (B_0, A))$ and $\hat{\Psi}(\tau'' \oplus (C_0, A))$ respectively. We clearly have that both g_0, h_0 extends g_0^τ , as both the computations $\hat{\Psi}(\tau' \oplus (B_0, A))$ and $\hat{\Psi}(\tau'' \oplus (C_0, A))$ extend $\hat{\Psi}(\tau \oplus (B_0 \upharpoonright t, A \upharpoonright t)) = g_0^\tau$. In particular, $g_0 \upharpoonright \text{dom}(g_0^\tau) = h_0 \upharpoonright \text{dom}(g_0^\tau)$.

By fixing $\gamma \in T_1 \upharpoonright t$ that is extendable to paths in both T_1 and $\hat{\Phi}(C_1, A)$, we may similarly obtain isomorphisms $g_1 : B_1 \rightarrow A$ and $h_1 : C_1 \rightarrow A$ such that $g_1 \upharpoonright \text{dom}(g_1^\gamma) = h_1 \upharpoonright \text{dom}(g_1^\gamma)$. Finally, fix some isomorphism $f : B_0 \rightarrow B_1$ extending f^σ and consider the isomorphism given by the composition below.

$$C_0 \xrightarrow{h_0} A \xrightarrow{g_0^{-1}} B_0 \xrightarrow{f} B_1 \xrightarrow{g_1} A \xrightarrow{h_1^{-1}} C_1.$$

We claim that $h_1^{-1} g_1 f g_0^{-1} h_0$ extends f^σ , equivalently, $h_1^{-1} g_1 f g_0^{-1} h_0 \upharpoonright \text{dom}(f^\sigma) = f^\sigma$. By property (4), we have that $\text{dom}(f^\sigma) \subseteq \text{dom}(g_0^\tau)$, and thus $g_0^{-1} h_0 \upharpoonright \text{dom}(f^\sigma)$ is the identity function on $\text{dom}(f^\sigma)$. That is,

$$h_1^{-1} g_1 f g_0^{-1} h_0 \upharpoonright \text{dom}(f^\sigma) = h_1^{-1} g_1 f \upharpoonright \text{dom}(f^\sigma).$$

Since $f \upharpoonright \text{dom}(f^\sigma) = f^\sigma$, we have that $\text{range}(f \upharpoonright \text{dom}(f^\sigma)) = \text{range}(f^\sigma)$. By property (6), this is contained in $\text{dom}(g_1^\gamma)$. But we also have that $g_1 \upharpoonright \text{dom}(g_1^\gamma) = h_1 \upharpoonright \text{dom}(g_1^\gamma)$, and thus $h_1^{-1} g_1 f \upharpoonright \text{dom}(f^\sigma) = f \upharpoonright \text{dom}(f^\sigma) = f^\sigma$.

We thus have that $\Psi(h_0 g_0^{-1} f g_1 h_1^{-1} \oplus (0^t 1^\mathbb{N}, 0^\mathbb{N})) = \Psi(f^\sigma \oplus (0^s, 0^s)) \downarrow = 0$, where $h_0 g_0^{-1} f g_1 h_1^{-1}$ is an isomorphism from C_0 to C_1 . However, C_0, C_1 are the pair of structures produced by $\Phi(0^t 1^\mathbb{N}, 0^\mathbb{N})$. Therefore, Φ, Ψ cannot possibly witness the reduction $\text{LLPO} \leq_w \text{cat}(\mathcal{A})$. $\dashv$

§5. Upper bounds of degrees for linear orders. In this section, we provide a largely self-contained approach to the analysis of the Weihrauch degrees of categoricity problems of various linear orders. This serves as some examples of a more general analysis that will be performed regarding Scott families and ranks in Section 6. First we recall the notion of a computable ordinal:

DEFINITION 5.1. [4, Chapter 4.6] The computable ordinals may be represented via *notations* (not unique in general), the collection of which we shall denote using $\mathcal{O} \subseteq \mathbb{N}$. n_α will be used to denote a notation for α . Now, together with $\mathcal{O}$, we also define an ordering $<_{\mathcal{O}}$ so that $n_\alpha <_{\mathcal{O}} n_\beta$ only if $\alpha < \beta$.

The (unique) notation for 0 is 1, and the remainder may be defined recursively as follows: If n_α is a notation for α , then 2^{n_α} is a notation for $\alpha + 1$. Evidently, we also define $n_\alpha <_{\mathcal{O}} 2^{n_\alpha}$. If e is such that for all n , $\varphi_e(n) \in \mathcal{O}$ and $\varphi_e(n) <_{\mathcal{O}} \varphi_e(n+1)$, then define $\varphi_e(n) <_{\mathcal{O}} 3(5^e)$ for all n , and let $3(5^e)$ be a notation for the least ordinal larger than each of the ordinals which $\varphi_e(n)$ represents.

DEFINITION 5.2. For a well order L let S_0 be the successor relation, B_0 be the relation consisting only of pairs (x, x) for each $x \in L$, and let m_0 be the trivial predicate which holds for all $x \in L$. For each computable ordinal $\alpha > 0$, define the following:

- $B_\alpha(x, y)$ is the reflexive and symmetric closure of the relation: “There exists $\beta_0, \beta_1, \dots, \beta_n < \alpha$ and $z_1, z_2, \dots, z_n$ so that $S_{\beta_0}(x, z_1), S_{\beta_1}(z_1, z_2), \dots, S_{\beta_n}(z_n, y)$.”
- $S_\alpha(x, y)$ is the relation: “ $x < y$, $\neg B_\alpha(x, y)$, and for every z such that $x < z < y$, $B_\alpha(x, z)$ or $B_\alpha(y, z)$ holds.”
- $m_\alpha(x)$ is the predicate: “For every $y < x$, $\neg B_\alpha(x, y)$ ”

For each of our predicates $P \in \{S, B, m\}$ above, we use $P_{<\alpha}$ to denote the collection of predicates $\{P_\beta \mid \beta < \alpha\}$ indexed by a given notation n_α for α . $P_{\leq \alpha}$ may be similarly defined.

LEMMA 5.3. *Let L be a linear order, α be a computable ordinal, and $x, y \in L$ be such that $B_\alpha(x, y)$. If z is such that $x < z < y$, then $B_\alpha(x, z)$. In particular, B_α is an equivalence relation.*

PROOF. First we argue that the stated property implies that B_α is an equivalence relation. By definition, B_α is symmetric and reflexive. It remains to argue that it is transitive. Let x, y, z be such that $B_\alpha(x, y)$ and $B_\alpha(y, z)$ holds. Without loss of generality, suppose that $x < y$. Clearly, if $y < z$, then we may ‘chain’ the two definitions $B_\alpha(x, y)$ and $B_\alpha(y, z)$ together to obtain that $B_\alpha(x, z)$. If $z < y$, then we have that $x < z < y$, and by the property stated in the lemma, we have that $B_\alpha(x, z)$.

We proceed by induction on α . The base case $\alpha = 0$ is trivial since there are no distinct x, y such that $B_0(x, y)$. Now suppose inductively that for all $\beta < \alpha$, $B_\beta(x, y)$ and $x < z < y$ implies that $B_\beta(x, z)$, and thus, B_β is an equivalence relation. Fix $x < z < y$ such that $B_\alpha(x, y)$ holds. By definition of B_α , we have that there exists $\beta_0, \beta_1, \dots, \beta_n < \alpha$ and $z_1, z_2, \dots, z_n$ so that $S_{\beta_0}(x, z_1), S_{\beta_1}(z_1, z_2), \dots, S_{\beta_n}(z_n, y)$. Note that we trivially have that $B_\alpha(x, z_i)$ for each z_i .

Since $x < z < y$, and L is a linear order, there must exist some $i \leq n$ so that $z_i \leq z \leq z_{i+1}$ (let $z_0 = x$ and $z_{n+1} = y$). For such an i , if $z = z_i$ or $z = z_{i+1}$, then we already have that $B_\alpha(x, z)$. Thus, we may suppose that $z_i < z < z_{i+1}$. By definition of S_{β_i} , either $B_{\beta_i}(z_i, z)$, or $B_{\beta_i}(z_{i+1}, z)$ holds. If $B_{\beta_i}(z_i, z)$ holds, then we obtain $B_\alpha(x, z)$ by applying the definition of B_{β_i} to z_i and z to extend our chain. If we instead have that $B_{\beta_i}(z_{i+1}, z)$, then we claim that $S_{\beta_i}(z_i, z)$ holds, thus giving $B_\alpha(x, z)$. Let w be such that $z_i < w < z$. In particular, $z_i < w < z_{i+1}$ and thus either $B_{\beta_i}(z_i, w)$ or $B_{\beta_i}(z_{i+1}, w)$ holds. By inductive hypothesis, we then obtain that $B_{\beta_i}(z_i, w)$ or $B_{\beta_i}(z, w)$, thus giving us $S_{\beta_i}(z_i, z)$ as desired. $\dashv$

Observe that Lemma 5.3 implies that S_α does not change B_α -equivalence classes. That is, if $S_\alpha(x, y)$ and $B_\alpha(x, w), B_\alpha(y, z)$ then $S_\alpha(w, z)$.

LEMMA 5.4. *Let L be a linear order, α be a computable ordinal, and $x, y \in L$ be such that $x < y$ and $B_\alpha(x, y)$ holds. Then there exists $z_1, z_2, \dots, z_n \in L$ and $\beta < \alpha$ such that $S_\beta(x, z_1), S_\beta(z_1, z_2), \dots, S_\beta(z_n, y)$. In fact, β may be chosen to be the maximum over the β_i given by the definition of $B_\alpha(x, y)$.*

PROOF. Let $x < y$ be given such that $B_\alpha(x, y)$, and let $z_1, z_2, \dots, z_n \in L$ and $\beta_0, \beta_1, \dots, \beta_n < \alpha$ be the witnesses to $B_\alpha(x, y)$. Take $\beta = \max_i \beta_i$. Then our task is to find $w_1, w_2, \dots, w_r$ such that

$$S_\beta(x, w_1), S_\beta(w_1, w_2), \dots, S_\beta(w_r, y).$$

Let i be the least such that $\beta_i < \beta$. If no such i exists, then taking $r = n$ and $w_i = z_i$ for each $i \leq n$ gives the desired conclusion. Now consider the following cases.

If there exists $j > i$ such that $\beta_j = \beta$, then pick the least such j . That is,

$$S_{\beta_i}(z_i, z_{i+1}), \dots, S_{\beta_{j-1}}(z_{j-1}, z_j), S_\beta(z_j, z_{j+1}),$$

where each of $\beta_i, \beta_{i+1}, \dots, \beta_{j-1} < \beta$. That is, we have that $B_\beta(z_i, z_j)$ and $z_i < z_j$ holds. We claim that $S_\beta(z_i, z_{j+1})$ also holds. Let z be such that $z_i < z < z_{j+1}$. By the definition of S_β and the assumption that $S_\beta(z_j, z_{j+1})$, if $z_j < z < z_{j+1}$, then either $B_\beta(z_j, z)$ or $B_\beta(z_{j+1}, z)$. Since B_β is an equivalence relation, this means that if $z_j < z < z_{j+1}$ either $B_\beta(z_i, z)$ or $B_\beta(z_{j+1}, z)$. On the other hand, if z is such that $z_i < z \leq z_j$, then we obtain from Lemma 5.3 that $B_\beta(z_i, z)$. Thus, $S_\beta(z_i, z_{j+1})$ holds as desired; we may replace the chain $S_{\beta_i}(z_i, z_{i+1}), \dots, S_{\beta_j}(z_j, z_{j+1})$ where $i < j$ with $S_\beta(z_i, z_{j+1})$.

If for each $j > i$, $\beta_j < \beta$, then note that $i > 0$, otherwise we would have that $\beta_i < \beta$ for all $i \leq n$, which cannot be as $\beta = \max_i \beta_i$. Additionally, by choice of i , we also have that $\beta_{i-1} = \beta$. That is, we have $S_\beta(z_{i-1}, z_i), S_{\beta_i}(z_i, z_{i+1}), \dots, S_{\beta_n}(z_n, y)$ and $\beta_i, \beta_{i+1}, \dots, \beta_n < \beta$. Then we obtain that $B_\beta(z_i, y)$ and $z_i < y$. Applying a similar argument as before allows one to conclude that $S_\beta(z_{i-1}, y)$. $\dashv$

DEFINITION 5.5. [4, Chapter 7.2] A computable Π_α formula of $L_{\omega_1, \omega}$ is coded by a 4-tuple $\langle \Pi, n_\alpha, \bar{x}, e \rangle$ representing the formula $\bigwedge \forall \bar{y} \psi$ where the conjunction is over (the codes of) $\langle \psi, \bar{y} \rangle \in W_e \cap P(\Sigma, n_\alpha)$. Here the set $P(\Sigma, n_\alpha)$ consists of the codes $\langle \Sigma, n_\beta, \bar{z}, e' \rangle$ where $n_\beta <_\mathcal{O} n_\alpha$. ($P(\Sigma, 1)$ consists of the Gödel numbers of quantifier-free formulas.)

Similarly, a computable Σ_α formula is coded by a 4-tuple $\langle \Sigma, n_\alpha, \bar{x}, e \rangle$ representing the formula $\bigvee \exists \bar{y} \psi$ where the disjunction is over (the codes of) $\langle \psi, \bar{y} \rangle \in W_e \cap P(\Pi, n_\alpha)$ and $P(\Pi, n_\alpha)$ consists of the 4-tuples $\langle \Pi, n_\beta, \bar{z}, e' \rangle$ where $n_\beta <_\mathcal{O} n_\alpha$. ($P(\Pi, 1) = P(\Sigma, 1)$.)

THEOREM 5.6. [4, Theorem 7.1]

1. Given a code for a computable Σ_α (or Π_α) formula φ , we can find a code for a computable Π_α (or Σ_α) formula ψ which is logically equivalent to $\neg\varphi$.
2. Given codes for a pair of computable Σ_α (or Π_α) formulas φ, ψ , we can find codes for computable Σ_α (or Π_α) formulas equivalent to $\varphi \vee \psi$ and $\varphi \wedge \psi$ respectively.
3. Given a code of a formula φ that is a finite Boolean combination of Σ_β and Π_β formulas, with codes using notations $n_\beta <_\mathcal{O} n_\alpha$ for a fixed n_α , we may find the codes of a computable Σ_α formula and a computable Π_α formula, with notation n_α , both logically equivalent to φ .
4. Given a code of a computable Σ_α formula φ , we can find the codes of a computable Σ_α formula logically equivalent to $\exists x \varphi$, and a computable $\Pi_{\alpha+1}$ formula equivalent to $\forall x \varphi$. This also holds for computable Π_α formulas, *mutatis mutandis*.

LEMMA 5.7. *For a given linear order L there exist computable functions f^B, f^S, f^m so that for any notation $n_\alpha \in \mathcal{O}$, $f^B(n_\alpha), f^S(n_\alpha)$, and $f^m(n_\alpha)$ produces codes for $B_\alpha(x, y), S_\alpha(x, y)$, and $m_\alpha(x)$ as $\Sigma_{2\alpha}, \Pi_{2\alpha+1}$, and $\Pi_{2\alpha+1}$ formulas respectively.*

PROOF. Let $f^B(1)$ be a code for equality, as $B_0(x, y)$ iff $x = y$. Then we evidently have that $f^B(1)$ is a Σ_0 formula, and consequently contained in $P(\Sigma, 2)$ (recall that 2 is the unique notation for 1). For $f^S(1)$, since $S_0(x, y)$ is of the form $\forall z\psi$ where ψ is a finite Boolean combination of quantifier-free formulas and B_0 , by applying Theorem 5.6, we obtain a Π_1 code for $S_0(x, y)$. $f^m(1)$ may be defined in a similar fashion.

Now fix some notation n_α , and let $n_{2\alpha}$ be the (unique) notation for 2α obtained from n_α . Assume inductively that for any notation $n_\beta <_{\mathcal{O}} n_\alpha$, $f^B(n_\beta)$ and $f^S(n_\beta)$ produce codes of a computable $\Sigma_{2\beta}$ formula and a computable $\Pi_{2\beta+1}$ formula equivalent to B_β and S_β respectively. Recall that these are respectively contained in $P(\Sigma, n_{2\beta+1})$ and $P(\Pi, n_{2\beta+2})$. Additionally, assume inductively that the codes for these formulas use notations $<_{\mathcal{O}} n_{2\alpha}$. (If $\beta < \alpha$ then $2\beta, 2\beta + 1 < 2\alpha$.)

From Definition 5.2, Lemmas 5.3 and 5.4, $B_\alpha(x, y)$ is equivalent to the countable (or finite) disjunction over $n_\beta <_{\mathcal{O}} n_\alpha$ of the formulas: “There exists $(z_i)_{i \leq n}$ so that $S_\beta(x, z_0), S_\beta(z_0, z_1), \dots, S_\beta(z_n, y)$ or $S_\beta(y, z_0), S_\beta(z_0, z_1), \dots, S_\beta(z_n, x)$.” By applying Theorem 5.6 and the inductive hypothesis, we obtain that $B_\alpha(x, y)$ is of the form $\bigvee_{n_\beta <_{\mathcal{O}} n_\alpha} \exists \bar{z} \psi_\beta$ where ψ_β is a computable $\Pi_{2\beta+1}$ formula, whose code uses notations $n_{2\beta+1} <_{\mathcal{O}} n_{2\alpha}$. By [4, Proposition 4.10], we have that $\{n_\beta \mid n_\beta <_{\mathcal{O}} n_\alpha\}$ is uniformly c.e. in n_α , and thus, we may obtain a $\Sigma_{2\alpha}$ code for B_α with notation $n_{2\alpha}$. Therefore, $f^B(n_\alpha) \in P(\Sigma, 2^{n_{2\alpha}})$ or equivalently $P(\Sigma, n_{2\alpha+1})$.

Now for $f^S(n_\alpha)$, we rewrite $S_\alpha(x, y)$ as $\forall z\psi$, where ψ is a finite Boolean combination of quantifier-free formulas and B_α . From $f^B(n_\alpha)$, and applying Theorem 5.6, we may then obtain a $\Pi_{2\alpha+1}$ code for ψ , and similarly for $\forall z\psi$. In particular, $f^S(n_\alpha) \in P(\Pi, n_{2\alpha+2})$ as desired. A similar argument works to show that a $\Pi_{2\alpha+1}$ code with notation $n_{2\alpha+1}$ may be obtained for m_α . $\dashv$

5.1. Well orders.

DEFINITION 5.8. For each computable ordinal $\alpha > 0$ and notation n_α , define $\text{TJ}_\alpha : 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ to be the function which maps $X \subseteq \mathbb{N}$ to $X^{(n_\alpha)} \subseteq \mathbb{N}$.

THEOREM 5.9. *For a linear order L and any notation n_α for a computable ordinal $\alpha \geq 1$,*

$$\text{cat}(L) \leq_W \begin{cases} \text{cat}(L, S_{<\alpha}, m_{<\alpha}) \star \text{TJ}_{2\alpha-1} & \text{if } \alpha < \omega, \\ \text{cat}(L, S_{<\alpha}, m_{<\alpha}) \star \text{TJ}_{2\alpha} & \text{otherwise.} \end{cases}$$

(Recall that the collections $P_{<\alpha}$ are indexed via a fixed ordinal notation for α .)

PROOF. Fix a notation n_α for α , then by Lemma 5.7, given any linear order L , there are computable functions f^S, f^m , which on input $n_\beta <_{\mathcal{O}} n_\alpha$, produces the $\Pi_{2\beta+1}$ codes with notations $n_{2\beta+1} <_{\mathcal{O}} n_{2\alpha}$ for S_β and m_β . From Theorem 5.6, we may further require that these are given as $\Delta_{2\alpha}^0$ codes since $2\beta + 1 < 2\alpha$ for $\beta < \alpha$. Finally, by applying [4, Theorem 7.5], we obtain that each of these formulas are $\Delta_{2\alpha}^0(L)$, which are respectively computable from $\text{TJ}_{2\alpha-1}(L)$ if $\alpha < \omega$ and $\text{TJ}_{2\alpha}(L)$ otherwise. That is, from any instance (L_0, L_1) of $\text{cat}(L)$ and a given notation n_α for $\alpha \geq 1$, we may produce using $\text{TJ}_{2\alpha-1}(L_0 \oplus L_1)$ (if $\alpha < \omega$) or $\text{TJ}_{2\alpha}(L_0 \oplus L_1)$ (otherwise) the collection of predicates $S_{<\alpha}^{L_0}, m_{<\alpha}^{L_0}, S_{<\alpha}^{L_1}$, and $m_{<\alpha}^{L_1}$. Then using $\text{cat}(L, S_{<\alpha}, m_{<\alpha})$, we may obtain an isomorphism from L_0 to L_1 as desired. $\dashv$

LEMMA 5.10. *Let L be a well order, $\alpha > 0$ be a computable ordinal, and $x, y \in L$ be such that $x < y$ and $B_\alpha(x, y)$ holds. Then there is a unique $\beta < \alpha$ and $z_1, z_2, \dots, z_n \in L$ such that $S_\beta(x, z_1), S_\beta(z_1, z_2), \dots, S_\beta(z_n, y)$ and $m_\beta(z_i)$ holds for each z_i .*

PROOF. Applying Lemma 5.4 allows us to obtain β and $z_1, z_2, \dots, z_n$ satisfying $S_\beta(x, z_1), S_\beta(z_1, z_2), \dots, S_\beta(z_n, y)$. It remains to replace each z_i with l_i for which $m_\beta(l_i)$ holds and show that this is unique.

For each z_i , we pick l_i to be the (unique) element so that $B_\beta(l_i, z_i)$ and $m_\beta(l_i)$ holds. Since L is a well order, each equivalence class must have a least element, so l_i must exist and $l_i \leq z_i$. Since S_β is consistent on B_β -equivalent elements we have that $S_\beta(x, l_1), S_\beta(l_1, l_2), \dots, S_\beta(l_n, y)$ as desired.

For uniqueness we can observe that this sequence shows that $B_{\beta+1}(x, y)$ but $\neg B_\beta(x, y)$. $\dashv$

THEOREM 5.11. *For a well order L and a computable ordinal α such that B_α has only finitely many equivalence classes, $\text{cat}(L, S_{<\alpha}, m_{<\alpha}) \leq_W \mathbb{C}_\mathbb{N}$.*

PROOF. Let L_0, L_1 , presentations of $(L, S_{<\alpha}, m_{<\alpha})$ be given. To compute an isomorphism from L_0 to L_1 , we first use $\mathbb{C}_\mathbb{N}$ to extract the least element in each distinct B_α -equivalence class. Such elements l may be characterised by the predicate $m_\alpha(l)$. Since there may only be finitely many such l and the predicate is co-c.e. given $S_{<\alpha}$, $\mathbb{C}_\mathbb{N}$ may produce the indices of such elements.

For each $i = 0, 1$, let $l_0^i < l_1^i < \dots < l_n^i$ be the elements satisfying $m_\alpha^{L_i}$. (Observe that these are distinct representatives of all equivalence classes under $B_\alpha^{L_i}$.) To define our isomorphism from $f : L_0 \rightarrow L_1$, we first map l_j^0 to l_j^1 for each $j \leq n$. Now fix some enumeration $\{\beta_k\}_{k \in \omega}$ of ordinals $< \alpha$. At each stage s , for each $k < s$, and for each x where $f(x)$ has already been defined, we search for $y > x$ such that both $m_{\beta_k}^{L_0}(y)$ and $S_{\beta_k}^{L_0}(x, y)$ holds. Similarly search for $z \in L_1$ for which both $m_{\beta_k}^{L_1}(z)$ and $S_{\beta_k}^{L_1}(f(x), z)$ holds, then define $f(y) = z$.

To verify that the construction works that f is a partial isomorphism we can use induction to show that each stage f maps each element to its correct place, (recall that there is only one isomorphism from L_0 to L_1). The base case of $l_i^0 \mapsto l_i^1$ is clear. For the inductive step we observe that relation given by $S_\beta(x, l) \wedge m_\beta(l)$ is a partial function by definition. Thus the way we add elements, does not map an element to the wrong place.

To show that f is defined on every element we use the concept of α -names defined as follows. For a computable ordinal $\beta \leq \alpha$, a β -name for an element $y \in L$ is a finite sequence of elements $\{y_{i,j}\}_{i \leq k, j \leq r_i}$ for $i \leq k$ and $j \leq r_i$, and a sequence of computable ordinals $\beta > \gamma_0 > \gamma_1 > \dots > \gamma_k$ with the following properties:

- $m_\beta(y_{0,0})$ holds.
- For each $i \leq k$ and $j \leq r_i$, $m_{\gamma_i}(y_{i,j})$ holds.
- For each $i \leq k$ and $j < r_i$, $S_{\gamma_i}(y_{i,j}, y_{i,j+1})$ holds.
- For each $i < k$, $S_{\gamma_{i+1}}(y_{i,r_i}, y_{i+1,0})$ holds.
- $y_{k,r_k} = y$.

We show via induction that for every computable ordinal $0 \leq \beta \leq \alpha$, and every element $y \in L$, y has a unique β -name. The base case $\beta = 0$ is trivial, every element is its own unique 0-name. Now suppose inductively that for all $\gamma < \beta$ and each $y \in L$, y has a unique γ -name.

Now let $y \in L$ be given. We construct a β -name for y as follows. Pick $y_{0,0}$ to be the unique element such that $B_\beta(y_{0,0}, y)$ and $m_\beta(y_{0,0})$ holds (note that $y_{0,0}$ exists because L is a well order). Applying Lemma 5.10 provides a unique $\gamma < \beta$ and $z_1, z_2, \dots, z_n$ so that $S_\gamma(y_{0,0}, z_1), S_\gamma(z_1, z_2), \dots, S_\gamma(z_n, y)$ where each z_i is such that $m_\gamma(z_i)$ holds. Now we further choose z_{n+1} to be the unique element satisfying $m_\gamma(z_{n+1})$ and $B_\gamma(z_{n+1}, y)$. Observe that $S_\gamma(z_n, z_{n+1})$ must hold as any z where $z_n < z < z_{n+1} \leq y$ is such that either $B_\gamma(z_n, z)$ or $B_\gamma(z_{n+1}, z)$ by the fact that $S_\gamma(z_n, y)$ holds. By picking $\gamma_0 = \gamma$, $y_{0,i} = z_i$ for each $0 < i \leq n+1$, and combining this with the γ -name for y , we obtain a β -name for y . In addition, since the γ -name is unique, and the choice of z_i are also unique, the β -name thus obtained for y must also be unique.

An α -name must start with $y_{0,0} = l_i$ for some i as these are the only elements for which m_α holds, so $y_{0,0} \in \text{dom}(f)$. By definition, we have that for each $j < r_i$, $S_{\gamma_i}(y_{i,j}, y_{i,j+1}) \wedge m_{\gamma_i}(y_{i,j+1})$ hold and for each $i < k$, $S_{\gamma_{i+1}}(y_{i,r_i}, y_{i+1,0}) \wedge m_{\gamma_{i+1}}(y_{i+1,0})$ holds. So by induction, we have that all of $\{y_{i,j}\}_{i \leq k, j \leq r_i}$ is in the domain of f , including $y = y_{k,r_k}$. Since every element has an α name, we have that f is defined everywhere, and so it is an isomorphism. $\dashv$

COROLLARY 5.12. *For a well order L and a computable ordinal α such that B_α has only finitely many equivalence classes:*

- $\text{cat}(L) \leq_W \mathbb{C}_\mathbb{N} \star \text{TJ}_{2\alpha-1}$ if $1 \leq \alpha < \omega$,
- $\text{cat}(L) \leq_W \mathbb{C}_\mathbb{N} \star \text{TJ}_{2\alpha}$ otherwise.

5.2. Scattered linear orders. We consider the following class of linear orders.

DEFINITION 5.13. Define the class $\mathcal{D}$ of linear orders as follows:

- $\mathcal{D}_0 = \{\emptyset, 1\}$,
- $\mathcal{D}_\alpha = \left\{ \sum_{i=0}^n \sum_{j \in \tau_i} L_{i,j} \mid L_{i,j} \in \bigcup_{\beta < \alpha} \mathcal{D}_\beta, \tau_i \in \{\omega, \omega^*, \zeta\}, n \in \omega \right\}$, and
- $\mathcal{D} = \bigcup_\alpha \mathcal{D}_\alpha$.

In contrast to well orders, linear orders in $\mathcal{D}$ are not as well behaved. Recall that one of the steps in extracting an isomorphism for well orders was to prove that every element has a unique α -name, but such a notion may not even be well-defined for a general element of $\mathcal{D}$. We adapt the approach of [26] to our setting and consider the following Weihrauch problem.

DEFINITION 5.14. Given two linear orders L_0, L_1 , define the predicates:

$$\text{Iso}_\alpha^{\leq}(x, y) = \begin{cases} 1 & \text{if } \{z \in L_0 \mid B_\alpha^{L_0}(x, z), z \leq_{L_0} x\} \cong \{z \in L_1 \mid B_\alpha^{L_1}(y, z), z \leq_{L_1} y\}, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\text{Iso}_\alpha^{\geq}(x, y) = \begin{cases} 1 & \text{if } \{z \in L_0 \mid B_\alpha^{L_0}(x, z), z \geq_{L_0} x\} \cong \{z \in L_1 \mid B_\alpha^{L_1}(y, z), z \geq_{L_1} y\}, \\ 0 & \text{otherwise.} \end{cases}$$

For convenience, we also denote the conjunction $\text{Iso}_\alpha^{\leq} \wedge \text{Iso}_\alpha^{\geq}$ as Iso_α . Just as in Definition 5.2, for a fixed notation n_α we define $\text{Iso}_{<\alpha}^{\leq}$, $\text{Iso}_{<\alpha}^{\geq}$, and $\text{Iso}_{<\alpha}$ as the collections of the respective formulas for $\beta < \alpha$.

Abusing notation, we also use $\text{Iso}_{<\alpha}(L_0, L_1, n_\alpha)$ to denote the Weihrauch problem which outputs the collection of predicates $\text{Iso}_{<\alpha}$ for L_0, L_1 indexed by the given notation n_α .

LEMMA 5.15. *There is a computable function g which on input n_α produces the $\Pi_{2\alpha+1}$ codes for $\text{Iso}_\alpha^{\leq}$ and $\text{Iso}_\alpha^{\geq}$.*

PROOF. We perform a similar analysis as in Lemma 5.7; observe that $\text{Iso}_\alpha^{\leq}(x, y)$ can be written as the countable (or finite if $\alpha < \omega$) conjunction over each $n_\beta <_{\mathcal{O}} n_\alpha$ of the formula: “For every $x_0, x_1, \dots, x_n \in L_0$, there exists $y_0, y_1, \dots, y_n \in L_1$ such that if $x_n = x$ and $\bigwedge_{i < n} S_\beta^{L_0}(x_i, x_{i+1})$, then $y_n = y$, $\bigwedge_{i < n} S_\beta^{L_1}(y_i, y_{i+1})$, and $\bigwedge_{i \leq n} \text{Iso}_\beta(x_i, y_i)$.” $\text{Iso}_\alpha^{\geq}(x, y)$ may be defined similarly by replacing the conditions $x_n = x$ and $y_n = y$ with $x_0 = x$ and $y_0 = y$ respectively.

Now we define g recursively and show by induction that for a notation n_α of $\alpha \geq 1$, $g(n_\alpha)$ produces the $\Pi_{2\alpha+1}$ codes with notation $n_{2\alpha+1}$ (obtained from n_α) of $\text{Iso}_\alpha^{\leq}$ and $\text{Iso}_\alpha^{\geq}$ as a tuple. For

$g(1)$ (recall that 1 is the notation for 0), produce a Π_0 code for tautology. (This can be effectively rewritten as a Π_1 code.) Let n_α be given and suppose that for any $n_\beta <_{\mathcal{O}} n_\alpha$, $g(n_\beta)$ produces the $\Pi_{2\beta+1}$ codes for $\text{Iso}_\beta^{\leq}$ and $\text{Iso}_\beta^{\geq}$ with notation $n_{2\beta+1}$ obtained from n_β . To define $g(n_\alpha)$, by Lemma 5.7, we may obtain for each $n_\beta <_{\mathcal{O}} n_\alpha$, the $\Pi_{2\beta+1}$ codes for S_β . By induction and Theorem 5.6, we may also obtain, using $g(n_\beta)$, the $\Pi_{2\beta+1}$ code for Iso_β with notation $n_{2\beta+1}$. Then, $\text{Iso}_\alpha^{\leq}$ (similarly $\text{Iso}_\alpha^{\geq}$) is of the form $\forall \bar{x} \exists \bar{y} \psi$ where ψ is a Boolean combination of $\Pi_{2\beta+1}$ (with codes using a fixed $n_{2\beta+1}$) and quantifier-free formulas. So by applying Theorem 5.6 yet again, a $\Sigma_{2\alpha}$ code for $\exists \bar{y} \psi$, and thus a $\Pi_{2\alpha+1}$ code for $\text{Iso}_\alpha^{\leq}$ may be obtained with notation $n_{2\alpha+1}$ extending n_α . $\dashv$

LEMMA 5.16. *For linear orders L_0, L_1 , a computable ordinal $\alpha \geq 1$, and any notation n_α for α ,*

$$\text{Iso}_{<\alpha}(L_0, L_1, n_\alpha) \leq_W \begin{cases} \text{TJ}_{2\alpha-1} & \text{if } \alpha < \omega, \\ \text{TJ}_{2\alpha} & \text{otherwise.} \end{cases}$$

PROOF. Applying Lemma 5.15, for any $n_\beta <_{\mathcal{O}} n_\alpha$, we may uniformly produce via g , the $\Pi_{2\beta+1}$ codes for Iso_β on L_0, L_1 with notation $n_{2\beta+1} <_{\mathcal{O}} n_{2\alpha}$. In particular, we may also produce the $\Delta_{2\alpha}$ codes for each Iso_β , which are each $\Delta_{2\alpha}^0(L_0 \oplus L_1)$. Then $\text{TJ}_{2\alpha-1}(L_0 \oplus L_1)$ (if $\alpha < \omega$) and $\text{TJ}_{2\alpha}(L_0 \oplus L_1)$ (otherwise) suffices to compute the predicates contained in $\text{Iso}_{<\alpha}$ for L_0, L_1 . $\dashv$

THEOREM 5.17. *For a computable ordinal α and $L \in \mathcal{D}_\alpha$,*

$$\text{cat}(L) \leq_W \begin{cases} \text{TJ}_{2\alpha} & \text{if } \alpha < \omega, \\ \text{TJ}_{2\alpha+1} & \text{otherwise.} \end{cases}$$

PROOF. We show that for a computable ordinal $\alpha \geq 1$ and $L \in \mathcal{D}_\alpha$,

$$\text{cat}(L) \leq_W \begin{cases} \text{TJ} \star (\text{Iso}_{<\alpha} \times \text{TJ}_{2\alpha-1}) & \text{if } \alpha < \omega, \\ \text{TJ} \star (\text{Iso}_{<\alpha} \times \text{TJ}_{2\alpha}) & \text{otherwise.} \end{cases}$$

Applying Lemma 5.16 then gives the desired result. Given L_0, L_1 presentations of L , we use $\text{TJ}_{2\alpha}(L_0 \oplus L_1)$, or $\text{TJ}_{2\alpha-1}(L_0 \oplus L_1)$ if $\alpha < \omega$, to produce the relations $B_{<\alpha}^{L_i}$ and $S_{<\alpha}^{L_i}$ for each $i < 2$. $\text{Iso}_\alpha(L_0, L_1, n_\alpha)$, also produces the collections of predicates $\text{Iso}_{<\alpha}^{\leq}$ and $\text{Iso}_{<\alpha}^{\geq}$ indexed by $n_\beta <_{\mathcal{O}} n_\alpha$.

Now define the isomorphism $f : L_0 \rightarrow L_1$ as follows. Non-uniformly fix the largest $n \in \mathbb{N}$ such that there exists $x_0, x_1, \dots, x_{n-1}$ where $B_\alpha^{L_0}(x_i, x_j)$ does not hold for any $i < j < n$. Since $L \in \mathcal{D}_\alpha$, such an n must exist. Recall that B_α is c.e. in $S_{<\alpha}$. Thus, given the relations $S_{<\alpha}^{L_i}$, we may produce in a co-c.e. manner, the indices of $x_0, x_1, \dots, x_{n-1} \in L_0$ and $y_0, y_1, \dots, y_{n-1} \in L_1$ so that they are each in distinct B_α blocks in their respective structures.

At stage s , let the current ‘guess’ of the representatives of the distinct B_α classes in L_0 and L_1 be $x_0, x_1, \dots, x_{n-1} \in L_0$ and $y_0, y_1, \dots, y_{n-1} \in L_1$. Define $f_s(x_i)$ to be the $<_{\mathbb{N}}$ -least element y so that $B_\alpha^{L_1}(y_i, y)$ and for the finitely many $n_\beta <_{\mathcal{O}} n_\alpha$ discovered by stage s , $\text{Iso}_\beta^{\leq}(x_i, y, n_\beta) = \text{Iso}_\beta^{\geq}(x_i, y, n_\beta) = 1$ (Recall that the collection of notations $n_\beta <_{\mathcal{O}} n_\alpha$ is uniformly c.e. in n_α).

Since there are only finitely many $B_\alpha^{L_1}$ classes and $B_\alpha^{L_1}$ is c.e. in $S_{<\alpha}^{L_1}$, for each $y \in L_1$, either we discover y_i so that $B_\alpha^{L_1}(y_i, y)$ in finite time, or we discover that $y_0, y_1, \dots, y_{n-1}$ are not in distinct $B_\alpha^{L_1}$ classes. In the latter case, define f_s to be the empty function and proceed to stage $s+1$ with a fresh ‘guess’ for $y_0, y_1, \dots, y_{n-1}$. If some x_i and x_j are discovered to be in the same $B_\alpha^{L_0}$ class, we also similarly define f_s to be the empty function and proceed to stage $s+1$ with a fresh ‘guess’ for $x_0, x_1, \dots, x_{n-1}$.

Now suppose that $x_0, x_1, \dots, x_{n-1} \in L_0$ and $y_0, y_1, \dots, y_{n-1} \in L_1$ currently look to be in different B_α classes in their respective structures and that we have succeeded in defining $f_s(x_i)$ for each

$i < n$. Arrange the $\beta < \alpha$ such that $n_\beta <_{\mathcal{O}} n_\alpha$ has been enumerated in ascending order as $\beta_0 < \beta_1 < \dots < \beta_{s-1}$. Initially set $D_s = \{x_0, x_1, \dots, x_{n-1}\}$ and do the following.

While there is some $x \in L_0 \upharpoonright s$ such that $x \notin D_s$: For the least $x \in L_0 \upharpoonright s$ such that $x \notin D_s$, let $x^* \in \text{dom}(f_s)$ and $j < s$ be such that $S_{\beta_j}^{L_0}(x^*, x)$ or $S_{\beta_j}^{L_0}(x, x^*)$, and for any $x^{**} \in \text{dom}(f_s)$, if $S_{\beta_l}^{L_0}(x^{**}, x)$ or $S_{\beta_l}^{L_0}(x, x^{**})$, then $j \leq l$. Intuitively, x^* is the ‘closest’ element currently in $\text{dom}(f_s)$ to x . If no such x^* exists, then add x into D_s .

Otherwise, search for $y \in L_1 \upharpoonright s$ but not in the range of f_s satisfying the following:

- $S_{\beta_j}^{L_1}(f(x^*), y)$ if $S_{\beta_j}^{L_0}(x^*, x)$, or $S_{\beta_j}^{L_1}(y, f(x^*))$ if $S_{\beta_j}^{L_0}(x, x^*)$.
- $\text{Iso}_{\beta_{s-1}}(x, y) = 1$.

Define $f(x) = y$ for the first such y found. If no such y is found, then leave $f_s(x) \uparrow$ and add x into D_s . On the other hand, if x is added into $\text{dom}(f_s)$, then remove from D_s each x' such that $x' \notin \text{dom}(f_s)$.

It is easy to see that D_s will eventually be equal $L_0 \upharpoonright s$ as the only time we remove elements from D_s is when we add a member of $L_0 \upharpoonright s$ to $\text{dom}(f_s)$. Furthermore, for each $s \in \mathbb{N}$, f_s is uniformly computable in $\text{Iso}_{<\alpha}^{\leq}, \text{Iso}_{>\alpha}^{\geq}, S_{<\alpha}^{L_0}$, and $S_{<\alpha}^{L_1}$. Now we verify that $f = \lim_s f_s$ exists, and is an isomorphism from L_0 to L_1 .

First, since $B_\alpha^{L_0}$ and $B_\alpha^{L_1}$ are c.e. in $S_{<\alpha}^{L_0}$ and $S_{<\alpha}^{L_1}$ respectively, the choice of $x_0, x_1, \dots, x_{n-1} \in L_0$ and $y_0, y_1, \dots, y_{n-1} \in L_1$ must stabilise. Similarly, for each y_i , there must be some $<_{\mathbb{N}}$ -least y where $B_\alpha^{L_1}(y_i, y)$ such that $\text{Iso}_\beta(x_i, y) = 1$ for each $\beta < \alpha$. Once discovered, $f_s(x_i)$ would have been defined to be y and never again changes. Additionally, such an $f_s \upharpoonright \{x_0, \dots, x_{n-1}\}$ is evidently extendable to an isomorphism.

Now for each $i \geq n$, let x_i be the $<_{\mathbb{N}}$ -least such that $S_\beta^{L_0}(x_i, x_j)$ or $S_\beta^{L_0}(x_j, x_i)$ for some $\beta < \alpha$ and $j < i$. Suppose inductively that for some $m \geq n - 1$, there exists some stage s_m such that for each $i \leq m$ and each $s \geq s_m$, $f_s(x_i) = f_{s_m}(x_i)$, and that $f_{s_m} \upharpoonright \{x_0, \dots, x_m\}$ is extendable to an isomorphism from L_0 to L_1 . By the choice of x_{m+1} , at a stage $s^* \geq s_m$ where $S_\beta^{L_0}(x_{m+1}, x_j)$ or $S_\beta^{L_0}(x_j, x_{m+1})$ is witnessed for some $\beta < \alpha$ and $j \leq m$, the construction always attempts to define $f_s(x_{m+1})$ after defining $f_s(x_m)$. Applying the inductive hypothesis then allows us to conclude that there must be some $y \in L_1$ for which we may define $f_s(x_{m+1}) = y$. In particular, any y satisfying $\text{Iso}_\gamma(x_{m+1}, y) = 1$ for each $\gamma < \alpha$, and such that $S_\beta^{L_1}(f_s(x_i), y)$ if $S_\beta^{L_0}(x_i, x_{m+1})$, or $S_\beta^{L_1}(y, f_s(x_i))$ if $S_\beta^{L_0}(x_{m+1}, x_i)$ suffices, where x_i is the ‘closest’ to x_{m+1} as described in the construction. Evidently, this also ensures that $f_s : \{x_0, x_1, \dots, x_{m+1}\} \rightarrow L_1 \upharpoonright s$ is extendable to an isomorphism. Take s_{m+1} to be the stage $s \geq s^* \geq s_m$ during which the $<_{\mathbb{N}}$ -least appropriate y has already entered L_1 . After such a stage, the construction always defines $f_s(x_{m+1}) = f_{s_{m+1}}(x_{m+1})$. It is also easy to see that every $x \in L_0$ is x_i for some i as x must be contained in the same $B_\alpha^{L_0}$ class as one of $x_0, x_1, \dots, x_{n-1}$. $\dashv$

§6. Relatively Δ_α^0 -categorical structures. Here we work only with computable relational languages L .

DEFINITION 6.1. [4, Chapter 6.6] Let $\mathcal{M}$ be a countable structure in a language L . A *Scott family* for the structure $\mathcal{M}$ is a countable family Θ of infinitary formulas in the language $L \cup \{\bar{c}\}$, where $\bar{c}$ is a finite tuple of parameters from $\mathcal{M}$, satisfying the following:

- for every tuple $\bar{a}$ from $\mathcal{M}$, there exists a formula $\theta(\bar{x}) \in \Theta$ such that $\mathcal{M} \models \theta(\bar{a})$,
- if $\theta(\bar{x}) \in \Theta$ and $\mathcal{M} \models \theta(\bar{a}) \wedge \theta(\bar{b})$ for some $\bar{a}, \bar{b}$ from $\mathcal{M}$, then there is an automorphism of $\mathcal{M}$ that takes $\bar{a}$ to $\bar{b}$.

If the tuple $\bar{c}$ is empty, then we say that the Scott family Θ is *parameter-free*.

It is known that relative Δ_α^0 -categoricity can be characterized in the terms of Scott families:

THEOREM 6.2 ([2, 11]). *Let α be a non-zero computable ordinal. A computable structure $\mathcal{M}$ is relatively Δ_α^0 -categorical if and only if $\mathcal{M}$ has a c.e. Scott family Θ consisting of computable Σ_α formulas.*

We will also need the following notion:

DEFINITION 6.3. [36, Chapter X] Let α be a non-zero computable ordinal. Recall that the set $P(\Sigma, n_\alpha)$ consists of codes $\langle \Sigma, n_\beta, \bar{z}, e' \rangle$ (where $n_\beta <_{\mathcal{O}} n_\alpha$) that represent computable Σ_β formulas, for $\beta < \alpha$. Note that we can view $P(\Sigma, n_\alpha)$ as a computable subset of $\mathbb{N}$ (such that its computable index is chosen uniformly in $n_\alpha \in \mathcal{O}$). As usual, we identify a computable Σ_β formula $\varphi(\bar{z})$ with its code $\langle \Sigma, n_\beta, \bar{z}, e' \rangle$.

The α -jump $\mathcal{M}^{(\alpha)}$ of a countable L -structure $\mathcal{M}$ is defined as follows.

- (a) Suppose that $\alpha < \omega$. Then the language of $\mathcal{M}^{(\alpha)}$ is equal to $L \cup \{P_\varphi\}_{\varphi \in P(\Sigma, 2^{n_\alpha})}$. (Recall that $2^{n_\alpha} = n_{\alpha+1}$ here). The restriction of $\mathcal{M}^{(\alpha)}$ to the language L is equal to $\mathcal{M}$, and

$$\mathcal{M}^{(\alpha)} \models P_\varphi(\bar{x}) \Leftrightarrow \mathcal{M} \models \varphi(\bar{x}).$$

In other words, for a natural number $k > 0$, the k -th jump $\mathcal{M}^{(k)}$ is the structure $\mathcal{M}$ enriched by all predicates definable by computable Σ_k formulas in $\mathcal{M}$. We will assume that $\mathcal{M}^{(0)} = \mathcal{M}$.

- (b) If $\alpha \geq \omega$, then $\mathcal{M}^{(\alpha)}$ is defined in the same way, except that now we restrict to the formulas $\varphi \in P(\Sigma, n_\alpha)$. For example, $\mathcal{M}^{(\omega)}$ is the structure $\mathcal{M}$ enriched by the predicates definable by computable Σ_t formulas, $t \in \omega$.

We establish some useful consequences of Theorem 6.2.

THEOREM 6.4. (1) *Every computable structure $\mathcal{M}$ satisfies*

$$\text{cat}(\mathcal{M}) \leq_w \begin{cases} \text{cat}(\mathcal{M}^{(\alpha-1)}) \star \text{TJ}_{\alpha-1}, & \text{if } 1 \leq \alpha < \omega, \\ \text{cat}(\mathcal{M}^{(\alpha)}) \star \text{TJ}_\alpha, & \text{if } \alpha \geq \omega. \end{cases}$$

- (2) *If the structure $\mathcal{M}$ is relatively Δ_α^0 -categorical, then we additionally have*

$$\mathcal{C}'_{\mathbb{N}} \geq_w \begin{cases} \text{cat}(\mathcal{M}^{(\alpha-1)}), & \text{if } 1 \leq \alpha < \omega, \\ \text{cat}(\mathcal{M}^{(\alpha)}), & \text{if } \alpha \geq \omega. \end{cases}$$

DEFINITION 6.5. For the sake of convenience, we introduce the following notation:

$$\mathcal{M}^{(\alpha)} = \begin{cases} \mathcal{M}^{(\alpha-1)}, & \text{if } 1 \leq \alpha < \omega, \\ \mathcal{M}^{(\alpha)}, & \text{if } \alpha \geq \omega. \end{cases}$$

We note that in Theorem 6.4 the jump $\mathcal{M}^{(\alpha)}$ does not necessarily have computable copies.

PROOF OF THEOREM 6.4. (1) By Theorem 5.6.3, we may assume that every additional predicate P_φ from the language of the structure $\mathcal{M}^{(\alpha)}$ is given as a Δ_α code (uniformly in φ). By applying [4, Theorem 7.5], we obtain that the predicates P_φ are $\Delta_\alpha^0(\mathcal{M})$, and hence, they are computable from $\text{TJ}_{\alpha-1}(\mathcal{M})$ if α is finite and $\text{TJ}_\alpha(\mathcal{M})$ if $\alpha \geq \omega$. So, given an instance $(\mathcal{N}_0, \mathcal{N}_1)$ of $\text{cat}(\mathcal{M})$ (and a fixed notation n_α of the ordinal α), by using

$$\begin{cases} \text{TJ}_{\alpha-1}(\mathcal{N}_0 \oplus \mathcal{N}_1), & \text{if } \alpha < \omega, \\ \text{TJ}_\alpha(\mathcal{N}_0 \oplus \mathcal{N}_1), & \text{if } \alpha \geq \omega, \end{cases}$$

we can uniformly compute all the predicates P_φ in each $\mathcal{N}_i$, and hence we compute $(\mathcal{N}_0^{(\alpha)}, \mathcal{N}_1^{(\alpha)})$ which is an instance of $\text{cat}(\mathcal{M}^{(\alpha)})$. Since the pairs $(\mathcal{N}_0, \mathcal{N}_1)$ and $(\mathcal{N}_0^{(\alpha)}, \mathcal{N}_1^{(\alpha)})$ have precisely the same isomorphisms f , we obtain the desired Weihrauch reduction.

(2) By Theorem 6.2, we fix a c.e. Scott family Θ containing computable Σ_α formulas $\theta(\bar{x}, \bar{c})$ with a fixed tuple of parameters $\bar{c} \in \mathcal{M}$. Consider an arbitrary formula $\theta \in \Theta$. By Theorem 5.6, we can (uniformly in θ) transform it into the following logically equivalent, computable formula:

$$\theta'(\bar{x}, \bar{c}) = \bigwedge_{j \in W} \exists \bar{y}_j \neg \varphi_j(\bar{y}_j, \bar{x}, \bar{c}),$$

where φ_j is a computable Σ_{β_j} formula for some $\beta_j < \alpha$. Inside the jump $\mathcal{M}^{(\alpha)}$, the formula θ' is logically equivalent to a computable Σ_1 formula

$$\theta^*(\bar{x}, \bar{c}) = \bigwedge_{j \in W} \exists \bar{y}_j \neg P_{\varphi_j}(\bar{y}_j, \bar{x}, \bar{c}).$$

Therefore, the family $\Theta^* = \{\theta^* : \theta \in \Theta\}$ is a c.e. Scott family for the structure $\mathcal{M}^{(\alpha)}$, and Θ^* contains only computable Σ_1 formulas.

Then a standard argument (see, e.g., [4, Theorem 10.14]) proves that the structure $\mathcal{M}^{(\alpha)}$ is relatively computably categorical (in the sense of Definition 1.2), i.e., for every instance $(\mathcal{A}_0, \mathcal{A}_1)$ of $\text{cat}(\mathcal{M}^{(\alpha)})$, there exists φ_e such that $\varphi_e^{(\mathcal{A}_0, \mathcal{A}_1)}$ is an isomorphism between $\mathcal{A}_0$ and $\mathcal{A}_1$. By repeating the proof of Theorem 2.2, then we show that $\text{cat}(\mathcal{M}^{(\alpha)}) \leq_W \mathbf{C}'_{\mathbb{N}}$. $\dashv$

We have the following corollary for parameter-free Scott families:

COROLLARY 6.6. *If a computable structure $\mathcal{M}$ has a parameter-free c.e. Scott family Θ consisting of computable Σ_α formulas, then*

$$\text{cat}(\mathcal{M}) \leq_W \begin{cases} \text{TJ}_{\alpha-1}, & \text{if } 1 \leq \alpha < \omega, \\ \text{TJ}_\alpha, & \text{if } \alpha \geq \omega. \end{cases}$$

PROOF. We analyze the proof of Theorem 6.4.(2) for this particular case. Here the c.e. Scott family Θ^* has no parameters $\bar{c}$, and hence an application of the construction from [4, Theorem 10.14] proves that $\text{cat}(\mathcal{M}^{(\alpha)}) \leq_W \text{id}$. $\dashv$

Theorem 6.4 only gives us an implication in one direction unlike Theorem 2.2. This leaves us to ask the following:

QUESTION 6.7. Does Theorem 6.4 reverse? Specifically is it the case that if

$$\text{cat}(\mathcal{M}) \leq_W \begin{cases} \mathbf{C}'_{\mathbb{N}} \star \text{TJ}_{\alpha-1}, & \text{if } 1 \leq \alpha < \omega, \\ \mathbf{C}'_{\mathbb{N}} \star \text{TJ}_\alpha, & \text{if } \alpha \geq \omega, \end{cases}$$

it must be that $\mathcal{M}$ is relatively Δ_α^0 -categorical?

Theorem 6.4 assumes a worst case on difficulty of finding the parameters of a Scott family, but for many structures, like the linear orders of Sections 3 and 5, the parameters are easier to find, only requiring us to appeal to $\mathbf{C}_{\mathbb{N}}$ rather than $\mathbf{C}'_{\mathbb{N}}$.

Scott families have a close connection to the Scott rank and Scott sentence complexity of a structure [36].

DEFINITION 6.8. For a countable structure, $\mathcal{M}$, we define the following:

- The (parameterized) Scott-rank of $\mathcal{M}$ is the least ordinal α such that $\mathcal{M}$ has a Scott family consisting of Σ_α formulas.
- The parameter free Scott-rank of $\mathcal{M}$ is the least ordinal α such that $\mathcal{M}$ has a parameter free Scott family consisting of Σ_α formulas.
- The Scott complexity of $\mathcal{M}$ is the complexity of the simplest Scott sentence for $\mathcal{M}$, one of of $\Sigma_\alpha, \Pi_\alpha, d\text{-}\Sigma_\alpha$ for some α .

THEOREM 6.9 ([36]). *The Scott-complexity, and Scott ranks of a structure are connected, with the only possible combinations coming from the examples in the table below.*

TABLE 1. Possible Scott complexities

Scott complexity	Scott rank	Parameter free Scott rank	Parameter complexity
$\Sigma_{\alpha+2}$	α	$\alpha + 2$	$\Pi_{\alpha+1}$
$d\text{-}\Sigma_{\alpha+1}$	α	$\alpha + 1$	Π_α
$\Pi_{\alpha+1}$	α	α	none
<i>When α is a limit</i>			
$\Sigma_{\alpha+1}$	α	$\alpha + 1$	Π_α
Π_α	α	α	none

The key observation from the above theorem is that the parameter complexity is always some Π_α sentence. If we know the complexity of the parameters, then we can give a better upper bound on $\text{cat}(\mathcal{M})$ than Theorem 6.4.

COROLLARY 6.10. *If M has an c.e. Scott family of computable Σ_α formulas with parameters defined by a computable Π_α formula, $\neg\psi(\bar{x})$, then:*

$$\text{cat}(\mathcal{M}) \leq_W \begin{cases} \mathbb{C}_\mathbb{N} \star \text{TJ}_{\alpha-1}, & \text{if } 1 \leq \alpha < \omega, \\ \mathbb{C}_\mathbb{N} \star \text{TJ}_\alpha, & \text{if } \alpha \geq \omega. \end{cases}$$

PROOF. From Theorem 6.4, it is enough for us to show that $\text{cat}(\mathcal{M}^{(\alpha)}) \leq_W \mathbb{C}_\mathbb{N}$. We consider the following problem **param** who's instance is a copy of $\mathcal{M}^{(\alpha)}$ and who's solution is a tuple $\bar{c}$ such that $\mathcal{M}^{(\alpha)} \models \neg\psi(\bar{c})$. $\bar{c}$ is only a solution if $\mathcal{M}^{(\alpha)} \not\models P_\psi(\bar{c})$, so clearly **param** $\leq_W \mathbb{C}_\mathbb{N}$. If we know the parameters, then, like in the proof of Corollary 6.6 we get a uniform construction of an isomorphism, so we have:

$$\text{cat}(\mathcal{M}) \leq_W \text{cat}(\mathcal{M}^{(\alpha)}) \star \text{TJ}_\alpha \leq_W (\text{param} \times \text{param}) \star \text{TJ}_\alpha \leq \mathbb{C}_\mathbb{N} \star \text{TJ}_\alpha$$

if $\alpha \geq \omega$, and a similar result if $\alpha < \omega$. ⊢

In this section we have been using computable infinitary formulas, but Table 1 is referring to arbitrary Scott sentences, not just computable ones. Since every countable structure has a Scott sentence, one can relativise the results in this section to get bounds on $\text{cat}(M)$ in the continuous Weihrauch degrees. The continuous Weihrauch degrees are the degrees of problems under reductions by continuous operators rather than just computable ones. We use $\equiv_{\text{cW}}$ for equivalent under continuous reductions.

Since the strong degree of categoricity stabilizes on a cone [16], we conjecture the following about the continuous Weihrauch degree of $\text{cat}(\mathcal{M})$

CONJECTURE 6.11. For a structure $\mathcal{M}$, the Scott rank of $\mathcal{M}$ and continuous Weihrauch degree of $\text{cat}(\mathcal{M})$ are closely related with the only possible combinations coming from the following table:

TABLE 2. Possible continuous Weihrauch degrees

Degree	Scott rank	Parameter free Scott rank	Parameter complexity
$\mathcal{C}'_{\mathbb{N}} \star \text{TJ}_{\alpha}$	α	$\alpha + 2$	$\Pi_{\alpha+1}$
$\mathcal{C}_{\mathbb{N}} \star \text{TJ}_{\alpha}$	α	$\alpha + 1$	Π_{α}
TJ_{α}	α	α	none
When α is a limit			
$\mathcal{C}_{\mathbb{N}} \star \text{TJ}_{\alpha}$	α	$\alpha + 1$	Π_{α}
TJ_{α}	α	α	none

Note that while TJ_{α} is only well defined as a Weihrauch degree when α is computable, it is always well defined as a continuous Weihrauch degree.

6.1. Weihrauch degrees for some familiar structures. First, we look at the ordinals $\alpha < \omega^{\omega}$.

PROPOSITION 6.12. *Suppose that $n \geq 1$ and $\beta < \omega^n$. Then $\text{cat}(\omega^n + \beta) \equiv_W \text{TJ}_{2n-1}$.*

PROOF. It is well-known (see, e.g., [4, Chapter 17.3]) that the structure $\alpha = \omega^n + \beta$ is relatively Δ_{2n}^0 -categorical.

Proof for $\leq_W \text{TJ}_{2n-1}$. We consider the ‘standard’ c.e. Scott family Θ for the structure α , this family consists of computable Σ_{2n} -formulas $\theta(\bar{x}, \bar{c})$ with a fixed tuple of parameters $\bar{c}$. Such a Scott family Θ could be explicitly recovered, say, from the analysis of Theorem 5.11 or [29, Section 2]. The key feature of the family Θ is the choice of parameters $\bar{c}$. If we look at the Cantor normal form

$$\alpha = \omega^n + \omega^{n_0} \cdot k_0 + \omega^{n_1} \cdot k_1 + \cdots + \omega^{n_t} \cdot k_t,$$

(where $n > n_0 > n_1 > \cdots > n_t$ and $0 < k_i < \omega$), then one chooses the parameters $\bar{c} = c_0 < c_1 < \cdots < c_{\ell}$ as follows:

- c_0 is the least element of α ,
- $c_1, c_2, \dots, c_{k_0}$ are the least elements of the successive ω^{n_0} -parts in the normal form,
- $c_{k_0+1}, c_{k_0+2}, \dots, c_{k_0+k_1}$ are the least elements of the successive ω^{n_1} -parts in the normal form, etc.

We show that *every* parameter c_i , $i \leq \ell$, is definable by a computable Σ_{2n} -formula in the structure α . Indeed, we can prove this fact by considering the paramers $c_{\ell}, c_{\ell-1}, \dots, c_1$ one by one (note that c_0 is always definable by a Π_1 -formula). The last parameter c_{ℓ} satisfies the Π_{2n_t+1} -formula

$$\eta_{\ell}(x) = m_{n_t}(x) \wedge \forall y(y > x \rightarrow B_{n_t}(x, y)),$$

and the previous parameter $c_{\ell-1}$ satisfies the following:

- if $k_t \geq 2$, then the Σ_{2n_t+2} -formula

$$\eta_{\ell-1}(x) = \exists y(x < y \wedge \eta_{\ell}(y) \wedge m_{n_t}(x) \wedge S_{n_t}(x, y)).$$

- if $k_t = 1$ and $t \geq 1$, then the $\Sigma_{2n_{t-1}+2}$ -formula

$$\eta_{\ell-1}(x) = \exists y(x < y \wedge \eta_{\ell}(y) \wedge m_{n_{t-1}}(x) \wedge (\forall z > x)(B_{n_{t-1}}(x, z) \vee B_{n_t}(y, z))).$$

If $t \geq 1$ and $k_{t-1} \geq 2$, then the parameter $c_{\ell-2}$ satisfies the $\Sigma_{2n_{t-1}+2}$ -formula

$$\eta_{\ell-2}(x) = \exists y(x < y \wedge \eta_{\ell-1}(y) \wedge m_{n_{t-1}}(x) \wedge S_{n_{t-1}}(x, y)),$$

et cetera.

Direct computations show that we will need at most Σ_{2n_0+2} -formulas to define every parameter c_i . Since $n_0 + 1 \leq n$, we conclude that every parameter is Σ_{2n} -definable. Therefore, by using the Σ_{2n} -definitions of $\bar{c}$, we can transform our Scott family Θ into a new c.e. *parameter-free* Scott family Θ^* for the structure α . By Corollary 6.6, we deduce that $\text{cat}(\alpha) \leq_W \text{TJ}_{2n-1}$.

Proof for $\geq_W \text{TJ}_{2n-1}$. The results of Ash (see Corollary 3.1 and Theorem 2a in [3]) imply the following: given a uniformly $\Delta_{2n-1}^0(X)$ -computable sequence of linear orders $(L_i)_{i \in \mathbb{N}}$ such that in each L_i , the number 0 is its least element, one can uniformly (in X and the sequence $(L_i)_{i \in \mathbb{N}}$) construct an X -computable sequence $(\mathcal{M}_i)_{i \in \mathbb{N}}$ such that $\mathcal{M}_i \cong \omega^{n-1} \cdot L_i$, and 0 is the least element of $\mathcal{M}_i$. We refer to, e.g., [37, Section 4] for a detailed analysis of uniformity in Ash's construction.

Given an oracle $X \subseteq \mathbb{N}$, we uniformly produce the following $\Delta_{2n-1}^0(X)$ -computable sequence:

$$L_i \cong \begin{cases} 1, & \text{if } i \notin X^{(2n-1)}, \\ 2, & \text{if } i \in X^{(2n-1)}. \end{cases}$$

By applying the Ash's construction, we can get a uniformly X -computable sequence

$$(1) \quad \mathcal{M}_i \cong \begin{cases} \omega^{n-1}, & \text{if } i \notin X^{(2n-1)}, \\ \omega^{n-1} \cdot 2, & \text{if } i \in X^{(2n-1)}. \end{cases}$$

Without loss of generality, we may assume that $c_i = \langle i, 0 \rangle$ is the least element in $\mathcal{M}_i$.

We fix computable copies $\mathcal{A} \cong \alpha$ and $\mathcal{B} \cong \beta$ such that all the relations $m_k(x), B_k(x, y), S_k(x, y)$, for $k \leq n$, are computable in both $\mathcal{A}$ and $\mathcal{B}$.

Then a computable operator Φ , given X , produces the following instance $(\mathcal{S}_0, \mathcal{S}_1)$ of $\text{cat}(\alpha)$:

$$\mathcal{S}_0 = \left(\sum_{i \in \mathbb{N}} \mathcal{M}_i \right) + \mathcal{B}, \quad \mathcal{S}_1 = \mathcal{A}.$$

Suppose that h is the unique isomorphism from $\mathcal{S}_0$ onto $\mathcal{S}_1$. Then we have

$$i \notin X^{(2n-1)} \Leftrightarrow \mathcal{M}_i \cong \omega^{n-1} \Leftrightarrow \mathcal{S}_0 \models S_{n-1}(c_i, c_{i+1}) \Leftrightarrow \mathcal{S}_1 \models S_{n-1}(h(c_i), h(c_{i+1})).$$

Therefore, one obtains a Turing operator Ψ such that $X^{(2n-1)} = \Psi^h$. The operators Φ and Ψ give the desired reduction $\text{TJ}_{2n-1} \leq_W \text{cat}(\alpha)$. Proposition 6.12 is proved. $\dashv$

PROPOSITION 6.13. *Suppose that $n \geq 1$, $M \geq 2$, and $\beta < \omega^n$. Then $\text{cat}(\omega^n \cdot M + \beta) \equiv_W \mathbb{C}_\mathbb{N} \star \text{TJ}_{2n-1}$.*

PROOF. Note that it is known that the order $\alpha = \omega^n \cdot M + \beta$ is relatively Δ_{2n}^0 -categorical.

For the sake of simplicity, here we give a proof for the case when $\alpha = \omega^n \cdot 2$. Details for the general case could be recovered similarly to the proof of Proposition 6.12.

Proof for $\leq_W \mathbb{C}_\mathbb{N} \star \text{TJ}_{2n-1}$. Consider the standard c.e. Scott family Θ for the structure $\alpha = \omega^n \cdot 2$. The family Θ consists of computable Σ_{2n} -formulas $\theta(\bar{x}, c_0, c_1)$, where c_0 is the least element of α , and c_1 is the first element of the 'upper' ω^n -part of α . Then the parameter tuple (c_0, c_1) is definable by a computable Π_{2n} formula

$$\eta(x_0, x_1) = (x_0 < x_1) \& \bigwedge_{i \in \{0,1\}} \neg \exists y S_{n-1}(y, x_i).$$

By Corollary 6.10, we obtain that $\text{cat}(\omega^n \cdot 2) \leq_W \mathbb{C}_\mathbb{N} \star \text{TJ}_{2n-1}$.

Proof for $\geq_W \mathbb{C}_\mathbb{N} \star \text{TJ}_{2n-1}$. Since $\mathbb{C}_\mathbb{N} \equiv_{sW} \lim_{\mathbb{N}}$, we shall prove a reduction $\lim_{\mathbb{N}} \star \text{TJ}_{2n-1} \leq_W \text{cat}(\omega^n \cdot 2)$.

Given an oracle $X \subseteq \mathbb{N}$ and a computable functional Γ , we have to compute not only the jump $X^{(2n-1)}$, but also the limit $a^* = \lim_{i \rightarrow \infty} \Gamma^{X^{(2n-1)}}(i)$ belonging to ω .

We construct an instance $(\mathcal{S}_0, \mathcal{S}_1)$ of $\text{cat}(\omega^n \cdot 2)$. We choose a computable copy $\mathcal{S}_1 \cong \omega^n \cdot 2$ such that all the relations $m_k(x), B_k(x, y), S_k(x, y)$, for $k \leq n$, are computable in $\mathcal{S}_1$.

Uniformly in X , we produce a $\Delta_{2n-1}^0(X)$ approximation $(Y_s)_{s \in \mathbb{N}}$ such that every Y_s is a finite set, $\emptyset \neq Y_s \subseteq Y_{s+1}$, and $X^{(2n-1)} = \bigcup_{s \in \mathbb{N}} Y_s$. For the sake of brevity, we will use the notation

$$Y = X^{(2n-1)}.$$

We describe an ancillary construction of a uniform $\Delta_{2n-1}^0(X)$ -computable sequence of linear orders $(R_{a,i,\ell})_{a,i,\ell \in \mathbb{N}}$. Informally speaking, the order $R_{a,i,\ell}$ ‘tracks down’ our ℓ -th attempt to test the condition $(\forall j \geq i)(\Gamma^Y(j) = a)$. Along the construction, we will define two ancillary functions:

- $g(a, i, s)$ which tracks down the relevant index ℓ for $R_{a,i,\ell}$ at a stage s ,
- $h(a, i, s)$ which ‘measures’ our current confidence in the condition $(\forall j \geq i)(\Gamma^Y(j) = a)$.

At a stage s , for given a and i , we are allowed to add new elements only into the order $R_{a,i,g(a,i,s)}$. As usual, we may assume that the value $\Gamma^{Y_s}(i)[s]$ is undefined for all $i > s$.

At stage 0, for all a, i, ℓ we define $R_{a,i,\ell}$ as an one-element linear order. We define $g(a, i, 0) = 0$ and $h(a, i, 0) = 0$ for all a, i .

At a stage $s + 1$, for each $i \leq s$ (one-by-one), we proceed as follows. If $\Gamma^{Y_s}(i)[s] \neq a$, then put $g(a, i, s + 1) = g(a, i, s) + 1$. In particular, this implies that the orders $R_{a,i,\ell}$, for $\ell \leq g(a, i, s)$, will never grow in the future.

If $\Gamma^{Y_s}(i)[s] \downarrow = a$, then we compute the following values:

- $j' = \max\{j \geq i : (\forall k)(i \leq k \leq j \rightarrow \Gamma^{Y_s}(k)[s] \downarrow = a)\}$,
- $p = (j' - i) + 1$.

Notice that $p \leq s + 1$. We define $h(a, i, s + 1) = p$ and $g(a, i, s + 1) = g(a, i, s)$.

If $h(a, i, s + 1) > \max\{h(a, i, t) : t \leq s\}$, then we add a new greatest element to the order $R_{a,i,g(a,i,s+1)}$, and we also set $g(a, j, s + 1) = g(a, j, s) + 1$ for all $j > i$. We say that the values $g(a, j, s)$, for $j > i$, were *refreshed* by the index i .

This concludes the construction of $R_{a,i,\ell}$. We observe the following properties of the construction:

- If $\Gamma^Y(i) \neq a$, then $\lim_s g(a, i, s) = \infty$ and all orders $R_{a,i,\ell}$, for $\ell \in \mathbb{N}$, are finite. Indeed, choose a stage s' such that $\Gamma^{Y_{s'}}(i)[s] \downarrow = \Gamma^Y(i)$ for all $s \geq s'$. Then the orders $R_{a,i,\ell}$ do not grow after the stage s' .
- If $\Gamma^Y(i) = a$ and there exists $j > i$ such that $\Gamma^Y(j) = b \neq a$, then all orders $R_{a,i,\ell}$, for $\ell \in \mathbb{N}$, are finite. Indeed, these orders stop growing after a stage s' such that $\Gamma^{Y_{s'}}(k)[s] \downarrow = \Gamma^Y(k)$ for all $s \geq s'$ and $k \leq j$.

Consequently, if $\Gamma^Y(i) \neq \lim_{j \rightarrow \infty} \Gamma^Y(j)$, then the orders $R_{a,i,\ell}$, for $i, \ell \in \mathbb{N}$, are finite.

Recall that $a^* = \lim_{i \rightarrow \infty} \Gamma^Y(i)$. Suppose that i^* is the least index such that $\Gamma^Y(i) = a^*$ for all $i \geq i^*$. Choose a stage s^* such that $\Gamma^{Y_s}(j)[s] \downarrow = \Gamma^Y(j)$ for all $s \geq s^*$ and $j \leq i^*$. Then after the stage s^* , the value $g(a^*, i^*, s)$ never changes (since for $j < i$, the value $h(\Gamma^Y(j), j, s)$ stops growing, and hence, this j cannot refresh $g(a^*, i^*, s)$). We define $\ell^* = \lim_s g(a^*, i^*, s)$. Since we have $\Gamma^Y(k) = a^*$ for all $k \geq i^*$, we deduce that

$$\limsup_{s \rightarrow \infty} h(a^*, i^*, s) = \infty,$$

and the order R_{a^*, i^*, ℓ^*} is isomorphic to ω .

If $k > i^*$, then $\Gamma^Y(k) = a^*$ and the value $g(a^*, k, s)$ will be refreshed infinitely often by the index i^* . Therefore, every order $R_{a^*, k, \ell}$ is finite.

We conclude that among the orders $R_{a, i, \ell}$, there exists a unique infinite order $R^* = R_{a^*, i^*, \ell^*}$. The order R^* is isomorphic to ω , and the first coordinate of its index is equal to the limit $a^* = \lim_{i \rightarrow \infty} \Gamma^{X^{(2n-1)}}(i)$.

Now we are ready to define the desired structure $\mathcal{S}_0$ from our instance $(\mathcal{S}_0, \mathcal{S}_1)$ of $\text{cat}(\omega^n \cdot 2)$. We introduce a uniform $\Delta_{2n-1}^0(X)$ -computable sequence of linear orders $(L_k)_{k \in \mathbb{N}}$:

$$L_{2\langle a, i, \ell \rangle} = R_{a, i, \ell}, \quad L_{2k+1} \cong \begin{cases} 1, & \text{if } k \notin X^{(2n-1)}, \\ 2, & \text{if } k \in X^{(2n-1)}. \end{cases}$$

Note that precisely one of the orders L_k is infinite: the order $L_{2\langle a^*, i^*, \ell^* \rangle}$ is isomorphic to ω . Therefore, the sum $\widehat{L} = \sum_{k \in \omega} L_k$ is isomorphic to $\omega \cdot 2$. In addition, the first element in the ‘upper’ ω -part of $\widehat{L}$ belongs to the order L_{2k_0+1} , where $L_{2k_0} \cong \omega$ and, hence, $k_0 = \langle a^*, i^*, \ell^* \rangle$.

Similarly to Proposition 6.12, by applying the construction of Ash, we get a uniformly X -computable sequence $\mathcal{M}_k \cong \omega^{n-1} \cdot L_k$, $k \in \omega$. We put

$$\mathcal{S}_0 = \sum_{k \in \mathbb{N}} \mathcal{M}_k.$$

It is clear that $\mathcal{S}_0$ is a copy of $\omega^n \cdot 2$.

Suppose that F is the unique isomorphism from $\mathcal{S}_0$ onto $\mathcal{S}_1$. An argument analogous to that of Proposition 6.12 shows that one can compute $X^{(2n-1)}$ from F . In addition, using F , we find the odd index $2k_0 + 1$ such that an element of the order $\mathcal{M}_{2k_0+1}$ is mapped by F to the first element in the ‘upper’ ω^n -part of $\mathcal{S}_1$. This k_0 is equal to $\langle a^*, i^*, \ell^* \rangle$, where $a^* = \lim_{i \rightarrow \infty} \Gamma^{X^{(2n-1)}}(i)$. Similarly to Proposition 6.12, we conclude that $\lim_{\mathbb{N}} \star \text{TJ}_{2n-1} \leq_W \text{cat}(\omega^n \cdot 2)$. Proposition 6.13 is proved. $\dashv$

We obtain a result similar to Proposition 6.12 for some familiar Boolean algebras. For a linear order L , by $\text{Int}(L)$ we denote the corresponding interval Boolean algebra.

PROPOSITION 6.14. *Suppose that α is a non-zero computable ordinal.*

(a) *Any α satisfies*

$$(2) \quad \text{cat}(\text{Int}(\omega^\alpha)) \leq_W \begin{cases} \text{TJ}_{2\alpha-1}, & \text{if } \alpha < \omega, \\ \text{TJ}_{2\alpha}, & \text{otherwise;} \end{cases} \quad \text{cat}(\text{Int}(\omega^\alpha \cdot (1 + \eta))) \leq_W \begin{cases} \text{TJ}_{2\alpha}, & \text{if } \alpha < \omega, \\ \text{TJ}_{2\alpha+1}, & \text{otherwise.} \end{cases}$$

(b) *If $\alpha = n$ is finite, then one can replace $\leq_W$ with $\equiv_W$ in Eq. (2).*

PROOF. Similar to the analysis of scattered linear orders, we can introduce the following computable infinitary formulas for Boolean algebras $\mathcal{B}$:

- $F_0(x)$ says that $x = 0$;
- $\text{Atom}_\alpha(x)$ states that $\neg F_\alpha(x) \wedge (\forall y <_{\mathcal{B}} x)(F_\alpha(y) \vee F_\alpha(x - y))$;
- $F_{\alpha+1}(x)$ says that $\bigvee_{\beta \leq \alpha} F_\beta(x) \vee \bigvee_{k \geq 0} \exists y_0 \dots \exists y_k (\bigwedge_{i \leq k} \text{Atom}_\alpha(y_i) \wedge x = y_0 \cup y_1 \cup \dots \cup y_k)$.
- For a limit ordinal λ , one defines $F_\lambda(x)$ as $\bigvee_{\beta < \lambda} F_\beta(x)$.

The set $F_\alpha(\mathcal{B})$ is sometimes called the α -th Fréchet ideal of $\mathcal{B}$. One can show that F_α is definable by a computable $\Sigma_{2\alpha}$ -formula, and Atom_α is definable by a computable $\Pi_{2\alpha+1}$ -formula (uniformly in a notation n_α for a given α). In addition, the following fact is known: $\mathcal{B} \models \text{Atom}_\alpha(x)$ if and

only if the corresponding relative algebra $\mathcal{B} \upharpoonright x$ (with domain $\{y : y \leq_{\mathcal{B}} x\}$) is isomorphic to the superatomic algebra $\text{Int}(\omega^\alpha)$.

(a) *Proof for $\leq_W$.* The superatomic Boolean algebra $\text{Int}(\omega^\alpha)$ is relatively $\Delta_{2\alpha}^0$ -categorical, and the algebra $\text{Int}(\omega^\alpha \cdot (1 + \eta))$ is relatively $\Delta_{2\alpha+1}^0$ -categorical. In addition, it is known that one can choose appropriate *parameter-free* Scott families for these structures (we refer to [4, Section 17.5] for the superatomic case, and to [5, Lemma 4] for the case of $\text{Int}(\omega^\alpha \cdot (1 + \eta))$). Hence, the desired Weihrauch reduction follows from Corollary 6.6.

(b) *Proof for $\geq_W$.* (1). We fix a computable copy $\mathcal{A}$ of the algebra $\text{Int}(\omega^n)$ such that all the relations $F_k(x), \text{Atom}_k(x)$, for $k < n$, are computable in $\mathcal{A}$.

Given an oracle $X \subseteq \mathbb{N}$, we construct the same uniformly X -computable sequence of linear orders $(\mathcal{M}_i)_{i \in \mathbb{N}}$ as in Eq. (1). Recall that c_i is the least element of $\mathcal{M}_i$. Inside the sum $\sum_{i \in \mathbb{N}} \mathcal{M}_i$, we will consider half-open intervals $J_i = [c_i, c_{i+1}]$.

Our computable operator Φ , given X , produces the following instance $(\mathcal{S}_0, \mathcal{S}_1)$ of $\text{cat}(\text{Int}(\omega^n))$:

$$\mathcal{S}_0 = \text{Int}\left(\sum_{i \in \mathbb{N}} \mathcal{M}_i\right), \quad \mathcal{S}_1 = \mathcal{A}.$$

Suppose that h is an arbitrary isomorphism from $\mathcal{S}_0$ onto $\mathcal{S}_1$. Then we have

$$i \notin X^{(2n-1)} \Leftrightarrow \mathcal{M}_i \cong \omega^{n-1} \Leftrightarrow \mathcal{S}_0 \models \text{Atom}_{n-1}(J_i) \Leftrightarrow \mathcal{S}_1 \models \text{Atom}_{n-1}(h(J_i)).$$

Therefore, one gets a Turing operator Ψ such that $X^{(2n-1)} = \Psi^h$. The operators Φ and Ψ give the desired reduction $\text{TJ}_{2n-1} \leq_W \text{cat}(\text{Int}(\omega^n))$.

(2) Again, we choose a computable $\mathcal{A} \cong \text{Int}(\omega^n \cdot (1 + \eta))$ such that the relations $F_k(x), \text{Atom}_k(x)$, for $k \leq n$, are computable in $\mathcal{A}$.

Given an oracle X , we uniformly produce the following $\Delta_{2n-1}^0(X)$ -computable sequence of linear orders:

$$L_i \cong \begin{cases} \omega \cdot (1 + \eta), & \text{if } i \notin X^{(2n)}, \\ \text{a finite ordinal } k_i, & \text{if } i \in X^{(2n)}. \end{cases}$$

By applying the Ash's construction, we get a uniformly X -computable sequence

$$\mathcal{M}_i \cong \begin{cases} \omega^n \cdot (1 + \eta), & \text{if } i \notin X^{(2n)}, \\ \omega^{n-1} \cdot k_i, & \text{if } i \in X^{(2n)}. \end{cases}$$

Our computable operator Φ , given X , produces the following instance $(\mathcal{S}_0, \mathcal{S}_1)$ of $\text{cat}(\text{Int}(\omega^n \cdot (1 + \eta)))$:

$$\mathcal{S}_0 = \sum_{i \in \mathbb{N}} \text{Int}(\mathcal{M}_i), \quad \mathcal{S}_1 = \mathcal{A}.$$

Let e_i be the greatest element in the direct summand $\text{Int}(\mathcal{M}_i)$ inside the algebra $\mathcal{S}_0$.

An arbitrary isomorphism h from $\mathcal{S}_0$ onto $\mathcal{S}_1$ satisfies:

$$i \in X^{(2n)} \Leftrightarrow \mathcal{M}_i \cong \omega^{n-1} \cdot k_i \Leftrightarrow \mathcal{S}_0 \models F_n(e_i) \Leftrightarrow \mathcal{S}_1 \models F_n(h(e_i)).$$

Therefore, we obtain a Turing operator Ψ such that $X^{(2n)} = \Psi^h$. We conclude that $\text{TJ}_{2n} \leq_W \text{cat}(\text{Int}(\omega^n \cdot (1 + \eta)))$. Proposition 6.14 is proved. $\dashv$

In conclusion of this section, we consider equivalence relations.

THEOREM 6.15. *Let U be the equivalence relation such that for each $n \in \omega \cup \{\omega\}$, U has infinitely many classes of size n . Then for any equivalence relation E , $\text{cat}(E) \leq_W \text{cat}(U)$.*

PROOF. Let $E_0, E_1 \cong E$ be given. Then define $U_0, U_1 \cong U$ as follows. Let U_1 be the ‘standard’ presentation of U ; a presentation such that given $m \in \mathbb{N}$, we can compute the size of the equivalence class containing m in U_1 . Let U_0 be the direct sum of $E_0, E_1, E_2 = U_1$. More formally, define for each $m \in \mathbb{N}$, $(3m + r, 3m' + r) \in U_0$ iff $(m, m') \in E_r$. It is easy to see that $U_0 \cong U_1$.

Now, given an isomorphism $f : U_0 \rightarrow U_1$, we define an isomorphism $g : E_0 \rightarrow E_1$ inductively as follows. Suppose that $g \upharpoonright m$ has been defined. To define $g(m)$, consider the following cases:

- If for each $m' < m$, $(m, m') \notin E_0$, then search for the least n such that n is not in the range of $g \upharpoonright m$ and such that $f(3m), f(3n + 1)$ are contained in equivalence classes of the same size (in U_1). Then define $g(m) = n$. From the assumption that $E_0 \cong E_1$, such a search must terminate.
- If there is some $m' < m$ such that $(m, m') \in E_0$, then search for the least n such that n is not in the range of $g \upharpoonright m$ and $(g(m'), n) \in E_1$, and define $g(m) = n$. It follows from the previous bullet point that such a search must terminate.

The map g as defined is evidently an isomorphism. $\dashv$

§7. Structural properties for problems $\text{cat}(\mathcal{M})$.

DEFINITION 7.1. By **CAT**₀ we denote the poset of Weihrauch degrees of all problems $\text{cat}(\mathcal{M})$, where $\mathcal{M}$ is a computable structure.

7.1. General connections to the degrees of categoricity. For an element $p \in \mathbb{N}^{\mathbb{N}}$, we fix the constant function

$$c_p : \mathbb{N}^{\mathbb{N}} \rightarrow \{p\}.$$

We note the following easy properties of the problems c_p :

- (A) $p \leq_T q$ if and only if $c_p \leq_W c_q$.
- (B) $c_p \leq_W \text{TJ}_\alpha$ if and only if $p \leq_T \emptyset^{(\alpha)}$.

We connect the problems c_p with the known literature on effective categoricity.

DEFINITION 7.2. [21] Let $\mathcal{M}$ be a computable structure. One says that a Turing degree $\mathbf{d}$ is the *strong degree of categoricity* for $\mathcal{M}$ if we have the following:

- (a) for any computable copies $\mathcal{M}_0 \cong \mathcal{M}_1 \cong \mathcal{M}$, there exists a $\mathbf{d}$ -computable isomorphism $f : \mathcal{M}_0 \rightarrow \mathcal{M}_1$,
- (b) there exist two computable copies $\mathcal{N}_0 \cong \mathcal{N}_1 \cong \mathcal{M}$ such that every isomorphism $g : \mathcal{N}_0 \rightarrow \mathcal{N}_1$ computes $\mathbf{d}$.

In this case, we also say that $\mathcal{M}$ has degree of categoricity $\mathbf{d}$ *strongly*.

There are many known examples of strong degrees of categoricity: for example, all degrees $\mathbf{d}$ such that $\mathbf{d} \leq_T \mathbf{0}'$ or $\mathbf{0}' \leq_T \mathbf{d} \leq_T \mathbf{0}''$ [14, 15]. We will use the following result:

THEOREM 7.3 ([15, Theorem 1]). *Let $\mathbf{d}$ be a Turing degree such that $\mathbf{d} \geq_T \mathbf{0}''$ and $\mathbf{d}$ is the degree of a Π_1^0 function singleton (i.e., there exists a computable tree $T \subset \mathbb{N}^{<\mathbb{N}}$ having a unique infinite path q , and this q satisfies $q \equiv_T \mathbf{d}$). Then one can build a rigid computable structure $\mathcal{M}$ having strong degree of categoricity $\mathbf{d}$.*

Note that there exist computable structures that have degrees of categoricity *non-strongly* [7, 17].

PROPOSITION 7.4. *Suppose that a computable structure $\mathcal{M}$ has strong degree of categoricity $\mathbf{d}$, and the automorphism group $\text{Aut}(\mathcal{M})$ is finite. Then $\mathbf{c}_p \leq_W \text{cat}(\mathcal{M})$ if and only if $p \leq_T \mathbf{d}$. Consequently, if $\mathbf{d} \neq \mathbf{0}^{(\alpha)}$, then $\text{cat}(\mathcal{M}) \not\equiv_W \text{TJ}_\alpha$.*

PROOF. ($\Rightarrow$). Suppose that $\mathbf{c}_p \leq_W \text{cat}(\mathcal{M})$, and this reducibility is witnessed by computable functionals $\Phi(x)$ and $\Psi(y, x)$. Consider computable copies $\mathcal{M}_0, \mathcal{M}_1$ produced as output of $\Phi(0^\infty)$. By Item (a) of Definition 7.2, we can choose an isomorphism $f: \mathcal{M}_0 \rightarrow \mathcal{M}_1$ such that $f \leq_T \mathbf{d}$. Then we have $p = \Psi(f, 0^\infty)$ and hence $p \leq_T f \leq_T \mathbf{d}$.

($\Leftarrow$). Assume that $p \leq_T \mathbf{d}$. Fix computable copies $\mathcal{N}_0, \mathcal{N}_1$ of $\mathcal{M}$ from Item (b) of Definition 7.2. Suppose that $h_i, i \leq k$, are all isomorphisms between $\mathcal{N}_0$ and $\mathcal{N}_1$. Notice that $p \leq_T \mathbf{d} \leq_T h_i$ for each i . We non-uniformly choose the following objects:

- Turing functionals $\Gamma_i, i \leq k$, such that $p = \Gamma_i^{h_i}$,
- a number $\ell \in \mathbb{N}$ such that $h_i \upharpoonright \ell \neq h_j \upharpoonright \ell$ for all $i < j \leq k$.

Then the desired Weihrauch reduction $\mathbf{c}_p \leq_W \text{cat}(\mathcal{M})$ is provided by the following operators Φ and Ψ : $\Phi(x)$ always outputs the pair $(\mathcal{N}_0, \mathcal{N}_1)$, and $\Psi(h, x)$ works as follows—if $h \upharpoonright \ell = h_i \upharpoonright \ell$, then $\Psi(h, x) = \Gamma_i^h$. $\dashv$

Proposition 7.4 allows us to separate the Weihrauch degrees of $\text{cat}(\mathcal{M})$ taken from Theorem 7.3:

PROPOSITION 7.5. *Suppose that $\mathcal{M}$ and $\mathcal{N}$ are computable structures such that $\mathcal{M}$ has strong degree of categoricity $\mathbf{d}_\mathcal{M}$, the automorphism group $\text{Aut}(\mathcal{M})$ is finite, and $\mathcal{N}$ has degree of categoricity $\mathbf{d}_\mathcal{N}$ (not necessarily strongly). If $\text{cat}(\mathcal{M}) \leq_W \text{cat}(\mathcal{N})$, then $\mathbf{d}_\mathcal{M} \leq_T \mathbf{d}_\mathcal{N}$. (Consequently, if $\mathbf{d}_\mathcal{M} \neq \mathbf{d}_\mathcal{N}$, then $\text{cat}(\mathcal{M}) \not\equiv_W \text{cat}(\mathcal{N})$).*

PROOF. Suppose that $\mathbf{d}_\mathcal{M} \not\leq_T \mathbf{d}_\mathcal{N}$. We choose $p \equiv_T \mathbf{d}_\mathcal{M}$. Then by Proposition 7.4, we have $\mathbf{c}_p \leq_W \text{cat}(\mathcal{M})$. On the other hand, the proof of ($\Rightarrow$) in Proposition 7.4 only relies on the assumption that one can choose a $\mathbf{d}$ -computable isomorphism f (we use neither the finiteness of the automorphism group, nor Item (b) of Definition 7.2). Hence, we deduce that $\mathbf{c}_p \not\leq_W \text{cat}(\mathcal{N})$. Therefore, we have $\text{cat}(\mathcal{M}) \not\leq_W \text{cat}(\mathcal{N})$. $\dashv$

The next natural question is the following: when can we ‘invert’ the implication from Proposition 7.5? That is, when $\mathbf{d}_\mathcal{M} \leq_T \mathbf{d}_\mathcal{N}$ implies $\text{cat}(\mathcal{M}) \leq_W \text{cat}(\mathcal{N})$?

The following general fact gives a partial answer to this question:

PROPOSITION 7.6. *Suppose that $\mathcal{M}$ and $\mathcal{N}$ are computable structures. There exists a computable structure $\mathcal{M} \oplus_P \mathcal{N}$ such that:*

1. *The automorphism group $\text{Aut}(\mathcal{M} \oplus_P \mathcal{N})$ is isomorphic to the direct product $\text{Aut}(\mathcal{M}) \times \text{Aut}(\mathcal{N})$.*
2. *We have $\text{cat}(\mathcal{M}) \sqcup \text{cat}(\mathcal{N}) \leq_W \text{cat}(\mathcal{M}) \times \text{cat}(\mathcal{N}) \equiv_W \text{cat}(\mathcal{M} \oplus_P \mathcal{N})$. In particular, the poset $\mathbf{CAT}_0$ is closed with respect to the product of problems $\times$.*
3. *If $\mathcal{M}$ and $\mathcal{N}$ have degrees of categoricity $\mathbf{d}_\mathcal{M}$ and $\mathbf{d}_\mathcal{N}$ respectively, then $\mathcal{M} \oplus_P \mathcal{N}$ has degree of categoricity $\mathbf{d}_\mathcal{M} \vee \mathbf{d}_\mathcal{N}$. In addition, if $\mathbf{d}_\mathcal{M} \leq_T \mathbf{d}_\mathcal{N}$ and $\mathcal{N}$ has its degree of categoricity $\mathbf{d}_\mathcal{N}$ strongly, then $\mathcal{M} \oplus_P \mathcal{N}$ also has degree of categoricity $\mathbf{d}_\mathcal{N}$ strongly.*

PROOF. Suppose that $\mathcal{M}$ and $\mathcal{N}$ have languages $L_\mathcal{M}$ and $L_\mathcal{N}$, respectively. Without loss of generality, we may assume that $L_\mathcal{M} \cap L_\mathcal{N} = \emptyset$. We choose a new unary predicate P , and consider a language $L = L_\mathcal{M} \cup L_\mathcal{N} \cup \{P\}$.

The desired L -structure $\mathcal{M} \oplus_P \mathcal{N}$ is essentially the disjoint union of $\mathcal{M}$ and $\mathcal{N}$: the $L_\mathcal{M}$ -substructure of $\mathcal{M} \oplus_P \mathcal{N}$ with domain $P^{\mathcal{M} \oplus_P \mathcal{N}}$ is isomorphic to $\mathcal{M}$, the $L_\mathcal{N}$ -substructure of $\mathcal{M} \oplus_P \mathcal{N}$ with domain $\neg P^{\mathcal{M} \oplus_P \mathcal{N}}$ is isomorphic to $\mathcal{N}$, and we do not add any new tuples to the relations R

from L —for example, if $R \in L_{\mathcal{M}}$ and $\mathcal{M} \oplus_P \mathcal{N} \models R(\bar{a})$, then all elements of $\bar{a}$ must belong to $P^{\mathcal{M} \oplus_P \mathcal{N}}$.

(1) It is immediate to see that $\text{Aut}(\mathcal{M} \oplus_P \mathcal{N}) \cong \text{Aut}(\mathcal{M}) \times \text{Aut}(\mathcal{N})$: indeed, every automorphism of $\mathcal{M} \oplus_P \mathcal{N}$ ‘splits’ into an automorphism of the P -part of $\mathcal{M} \oplus_P \mathcal{N}$ (this part is essentially a copy of $\mathcal{M}$) and an automorphism of the $\neg P$ -part (a copy of $\mathcal{N}$).

(2) Since both problems $\text{cat}(\mathcal{M})$ and $\text{cat}(\mathcal{N})$ have computable instances, we deduce that $\text{cat}(\mathcal{M}) \sqcup \text{cat}(\mathcal{N}) \leq_W \text{cat}(\mathcal{M}) \times \text{cat}(\mathcal{N})$ (see, e.g., Proposition 11.5.7 in [8]).

The equivalence $\text{cat}(\mathcal{M}) \times \text{cat}(\mathcal{N}) \equiv_W \text{cat}(\mathcal{M} \oplus_P \mathcal{N})$ is proved in a straightforward way. Indeed, observe that one can introduce the following computable operators Γ and Δ : given copies $\mathcal{M}_0 \cong \mathcal{M}$ and $\mathcal{N}_0 \cong \mathcal{N}$, Γ outputs a copy of $\mathcal{M} \oplus_P \mathcal{N}$ by copying $\mathcal{M}_0$ to the even numbers and putting $\mathcal{N}_0$ to the odd numbers. Given a copy $\mathcal{S} \cong \mathcal{M} \oplus_P \mathcal{N}$, Δ produces a copy of $\mathcal{M}$ and a copy of $\mathcal{N}$ by splitting $\mathcal{S}$ into its P -part and $\neg P$ -part. By using the operators Γ, Δ and the fact that automorphisms of $\mathcal{M} \oplus_P \mathcal{N}$ also ‘split’ nicely, one can easily recover the desired Weihrauch equivalence.

(3) Suppose that $\mathcal{M}$ and $\mathcal{N}$ have degrees of categoricity $\mathbf{d}_{\mathcal{M}}$ and $\mathbf{d}_{\mathcal{N}}$, respectively. It is clear that every degree $\mathbf{d} \geq_T \mathbf{d}_{\mathcal{M}} \vee \mathbf{d}_{\mathcal{N}}$ can compute isomorphisms between arbitrary computable copies of $\mathcal{M} \oplus_P \mathcal{N}$.

Fix a computable copy $\mathcal{M}_0 \cong \mathcal{M}$. If $\mathbf{c} \not\geq_T \mathbf{d}_{\mathcal{M}} \vee \mathbf{d}_{\mathcal{N}}$, then either $\mathbf{c} \not\geq_T \mathbf{d}_{\mathcal{M}}$ or $\mathbf{c} \not\geq_T \mathbf{d}_{\mathcal{N}}$. We may assume that $\mathbf{c} \not\geq_T \mathbf{d}_{\mathcal{N}}$. Then there exist computable copies $\mathcal{A} \cong \mathcal{B} \cong \mathcal{N}$ such that there is no $\mathbf{c}$ -computable isomorphism from $\mathcal{A}$ onto $\mathcal{B}$. Hence, $\mathbf{c}$ also cannot compute an isomorphism between computable structures $\mathcal{M}_0 \oplus_P \mathcal{A}$ and $\mathcal{M}_0 \oplus_P \mathcal{B}$. We conclude that $\mathbf{d}_{\mathcal{M}} \vee \mathbf{d}_{\mathcal{N}}$ is the degree of categoricity for $\mathcal{M} \oplus_P \mathcal{N}$.

Now assume that $\mathbf{d}_{\mathcal{M}} \leq_T \mathbf{d}_{\mathcal{N}}$, and that $\mathbf{d}_{\mathcal{N}}$ is the strong degree of categoricity for $\mathcal{N}$. We fix computable copies $\mathcal{A} \cong \mathcal{B} \cong \mathcal{N}$ witnessing Item (b) of Definition 7.2 for $\mathcal{N}$. Then every isomorphism g from $\mathcal{M}_0 \oplus_P \mathcal{A}$ onto $\mathcal{M}_0 \oplus_P \mathcal{B}$ computes some isomorphism f from $\mathcal{A}$ onto $\mathcal{B}$, and thus, $g \geq_T f \geq_T \mathbf{d}_{\mathcal{N}}$. We deduce that $\mathcal{M} \oplus_P \mathcal{N}$ has its degree of categoricity $\mathbf{d}_{\mathcal{N}}$ strongly. $\dashv$

COROLLARY 7.7. *The poset $\mathbf{CAT}_0$ is closed with respect to parallelization of problems.*

PROOF. Similarly to Proposition 7.6, given a computable L -structure $\mathcal{M}$ we choose unary predicates $P_i \notin L$, $i \in \mathbb{N}$, and we define a new computable structure $\widehat{\mathcal{M}}$ as follows. The structure $\widehat{\mathcal{M}}$ is a disjoint union of L -structures $\mathcal{M}_i \cong \mathcal{M}$, for $i \in \mathbb{N}$, and a predicate P_i is true only on the elements from $\mathcal{M}_i$. It is straightforward to show that $\text{cat}(\widehat{\mathcal{M}})$ is Weihrauch equivalent to the parallelization $\widehat{\text{cat}(\mathcal{M})}$. $\dashv$

Now we are ready to give our first embedding into $\mathbf{CAT}_0$, i.e., the collection of all $\text{cat}(\mathcal{M})$.

COROLLARY 7.8. *Every finite poset is embeddable into $\mathbf{CAT}_0$. Consequently, the $\exists$ -theory of $(\mathbf{CAT}_0, \leq_W)$ is decidable.*

PROOF. It is known that every finite poset is embeddable into the interval $[\mathbf{0}'', \mathbf{0}''']$ of Turing degrees, and that every degree $\mathbf{d}$ in this interval contains a Π_1^0 function singleton. Hence, it is sufficient to embed into $\mathbf{CAT}_0$ any finite set of Turing degrees $\{\mathbf{d}_0, \dots, \mathbf{d}_n\} \subset [\mathbf{0}'', \mathbf{0}''']$. By Theorem 7.3, for each $\mathbf{d}_i$, $i \leq n$, we can choose a rigid computable structure $\mathcal{M}_i$ having degree of categoricity $\mathbf{d}_i$ strongly.

We define a new family of rigid computable structures $(\mathcal{N}_i)_{i \leq n}$ such that $\mathbf{d}_i$ is the strong degree of categoricity for $\mathcal{N}_i$. We put $\mathcal{N}_0 = \mathcal{M}_0$. Suppose that the structures $\mathcal{N}_0, \mathcal{N}_1, \dots, \mathcal{N}_i$ are already defined, and choose all the indices $j_0 < j_1 < \dots < j_t \leq i$ satisfying $\mathbf{d}_{j_k} <_T \mathbf{d}_{i+1}$. We iteratively apply Proposition 7.6 and construct

$$\mathcal{N}_{i+1} = (\dots ((\mathcal{M}_{i+1} \oplus_P \mathcal{N}_{j_0}) \oplus_P \mathcal{N}_{j_1}) \dots) \oplus_P \mathcal{N}_{j_t}.$$

Note that Proposition 7.6.2 implies that $\text{cat}(\mathcal{N}_{j_k}) \leq_W \text{cat}(\mathcal{N}_{i+1})$ for all $k \leq t$.

By Proposition 7.5, $\text{cat}(\mathcal{N}_i) \leq_W \text{cat}(\mathcal{N}_j)$ implies $\mathbf{d}_i \leq_T \mathbf{d}_j$. Therefore, we deduce that the map $\mathbf{d}_i \mapsto \text{cat}(\mathcal{N}_i)$ is an isomorphic embedding of posets. $\dashv$

We leave the following questions open.

QUESTION 7.9. Which familiar operations (except product and parallelization) is the poset $\mathbf{CAT}_0$ closed under?

QUESTION 7.10. What is the complexity of the first-order theory $\text{Th}(\mathbf{CAT}_0, \leq_W)$?

7.2. Embedding countable posets into $\mathbf{CAT}_0$. Our next result shows that one can push the embedding of Corollary 7.8 even further:

THEOREM 7.11. *For every c.e. Turing degree $\mathbf{d}$, there exists a computable structure G such that $\text{cat}(G) \equiv_W \mathbf{c}_p$ for $p \equiv_T \mathbf{d}$. Consequently, the upper semilattice of c.e. Turing degrees is embeddable (as a poset) into $\mathbf{CAT}_0 \cap \{\mathbf{P} : \mathbf{P} \leq_W \mathbf{TJ}\}$.*

Theorem 7.11 and a result of Sacks [40] together imply:

COROLLARY 7.12. *Every countable poset is embeddable into $\mathbf{CAT}_0 \cap \{\mathbf{P} : \mathbf{P} \leq_W \mathbf{TJ}\}$.*

PROOF OF THEOREM 7.11. We use an idea from [21]. Given a c.e. set V , we construct a computable directed irreflexive graph $G(V)$ as follows. For each $i \in \mathbb{N}$, $G(V)$ contains a unique directed simple cycle C_i of size $i + 3$. We choose a node $a_i \in C_i$ and two fresh nodes b_i and d_i . We connect additional (fresh) chains as follows:

$$\left\{ \begin{array}{l} a_i \rightarrow b_i \rightarrow \circ, \\ a_i \rightarrow d_i \rightarrow \circ \rightarrow \circ \end{array} \right\}, \text{ if } i \notin V; \quad \left\{ \begin{array}{l} a_i \rightarrow b_i \rightarrow \circ \rightarrow \circ, \\ a_i \rightarrow d_i \rightarrow \circ \rightarrow \circ \rightarrow \circ \end{array} \right\}, \text{ if } i \in V.$$

We collect some simple properties of the graph $G(V)$:

- (a) The graph $G(V)$ is constructed uniformly in a c.e. index of the set V , and $G(V)$ is rigid.
- (b) There exists a single Turing functional Ψ such that given an instance (H_0, H_1) of $\text{cat}(G(V))$, the map $h = \Psi^{H_0 \oplus H_1 \oplus V}$ is the unique isomorphism from H_0 onto H_1 . Indeed, knowing the set V allows one to effectively find the lengths of the chains connected to b_i and d_i , for $i \in \mathbb{N}$. Using these lengths, one can uniformly recover the isomorphism h . Therefore, we have $\text{cat}(G(V)) \leq_W \mathbf{c}_{\chi_V}$, where χ_V is the characteristic function of V .
- (c) The graph $G(V)$ has strong degree of categoricity $\deg_T(V)$. Indeed, by switching the lengths of the chains (see Eq. (3) below), one can build a computable graph $G_1(V)$ such that the unique isomorphism $f: G(V) \rightarrow G_1(V)$ computes V . By Proposition 7.4, we deduce that $\mathbf{c}_{\chi_V} \leq_W \text{cat}(G(V))$.

$$(3) \quad \left\{ \begin{array}{l} a_i \rightarrow b'_i \rightarrow \circ, \\ a_i \rightarrow d'_i \rightarrow \circ \rightarrow \circ \end{array} \right\}, \text{ if } i \notin V; \quad \left\{ \begin{array}{l} a_i \rightarrow b'_i \rightarrow \circ \rightarrow \circ \rightarrow \circ, \\ a_i \rightarrow d'_i \rightarrow \circ \rightarrow \circ \end{array} \right\}, \text{ if } i \in V.$$

We conclude that $\text{cat}(G(V)) \equiv_W \mathbf{c}_{\chi_V}$. Furthermore, since V is c.e., we have $V \leq_T \mathbf{0}'$ and $\mathbf{c}_{\chi_V} \leq_W \mathbf{TJ}$. Therefore, the map $V \mapsto \text{cat}(G(V))$ induces an embedding of the semilattice of c.e. Turing degrees into $\mathbf{CAT}_0 \cap \{\mathbf{P} : \mathbf{P} \leq_W \mathbf{TJ}\}$. Theorem 7.11 is proved. $\dashv$

In Section 10 we will show that $\mathbf{CAT}_0$ has a maximal element, with Theorem 10.1. The proof of this requires the universal tree presented in Section 8.

§8. The general problem. In this section we study the general **cat** problem and its place in the Weihrauch degrees as well as its relationship to $\text{cat}(\mathcal{M})$. The main result we are working towards is the following.

THEOREM 8.1. $\mathbb{C}_{\mathbb{N}} \equiv_{\text{W}} \text{cat}$.

To do this, we will want to use the notion of the universal tree. Rather than thinking of trees as subsets of $\mathbb{N}^{<\mathbb{N}}$ we will think of them as structures in the language of directed graphs $\{D\}$. We can think of $D(n, m)$ as meaning that n is the child of m in a tree and write $D(n) = m$. In this language, we can define a tree as a structure on which D is a function whose domain consists of every node except the root and every node is a finite distance from the root. We will use $\text{root}(T)$ to denote the unique root node of a tree T .

DEFINITION 8.2. Given a tree, T , we can define a rank function on the nodes of T using transfinite recursion. For $\sigma \in T$ we define $\text{rank}(\sigma)$ to be:

- 0 if σ is not in the image of D .
- $\sup\{\text{rank}(\tau) : D(\tau) = \sigma\}$ if rank is defined on all of $D^{-1}(\sigma)$.
- ∞ if rank can not otherwise be defined on σ .

This definition matches up with the standard definition we use on subsets of $\mathbb{N}^{<\mathbb{N}}$ and has that the rank of the root is ∞ if and only if the tree is ill-founded. Now we can define the universal tree.

DEFINITION 8.3. A tree, T , has the *universal property* if the rank of the root is ∞ and for each node $\sigma \in T$ the following hold:

- For each $\beta < \min(\text{rank}(\sigma, \omega_1^{CK}))$, σ has ω many children τ such that $\text{rank}(\tau) = \beta$.
- $\text{rank}(\sigma) < \omega_1^{CK}$ or $\text{rank}(\sigma) = \infty$.
- If $\text{rank}(\sigma) = \infty$, then σ has ω many children τ such that $\text{rank}(\tau) = \infty$.

This definition can be relativized to an oracle X , replacing ω_1^{CK} with ω_1^X . In this case we say the tree has the *X-universal property*.

We now justify why *the* universal tree is unique up to isomorphism.

LEMMA 8.4. *If T and S had the universal property then $T \cong S$.*

PROOF. We can do a back and forth construction starting at the root by ensuring that each new node is the child of some node used before and ensuring that each node is mapped to a node of the same rank. We will always be able to do this as each node in a universal tree has the maximum number of children of rank β it can for each $\beta \in \omega_1^{CK} \cup \{\infty\}$. $\dashv$

The plan now is to show that given any ill-founded tree T , we can ‘homogenise’ T (see Lemma 8.7) by taking its product (as defined below) with the universal tree relative to T .

DEFINITION 8.5. For trees T, S we define $T \times S$ as the tree whose domain consists of all pairs of nodes at the same height:

$$\{\langle \sigma, \tau \rangle : \sigma \in T, \tau \in S \wedge \exists n [D^n(\sigma) = \text{root}(T) \wedge D^n(\tau) = \text{root}(S)]\}.$$

We can then define $D(\langle \sigma, \tau \rangle) = \langle D(\sigma), D(\tau) \rangle$.

LEMMA 8.6. *For a product tree $T \times S$ we have that $\text{rank}(\langle \sigma, \tau \rangle) = \min(\text{rank}(\sigma), \text{rank}(\tau))$.*

PROOF. Transfinite induction on $\min(\text{rank}(\sigma), \text{rank}(\tau))$. $\dashv$

LEMMA 8.7. *If T is a computable tree of rank ∞ and $\mathcal{U}$ is the universal tree, then $T \times \mathcal{U} \cong \mathcal{U}$.*

PROOF. By Lemma 8.4 it is sufficient for us to show that $T \times \mathcal{U}$ has the universal property. Since T is computable we can observe that if a node in T is of ordinal rank, then it is of computable rank. By Lemma 8.6 we know that the root of $T \times \mathcal{U}$ will have rank ∞ and that every node of ordinal rank must have rank $< \omega_1^{CK}$.

Consider some node $\langle \sigma, \tau \rangle$ and some $\beta < \min(\text{rank}(\sigma), \text{rank}(\tau), \omega_1^{CK})$ (or $\beta = \infty = \text{rank}(\sigma) = \text{rank}(\tau)$). Since $\beta < \text{rank}(\tau)$ (or $\beta = \infty = \text{rank}(\tau)$) there must be some child, ρ , of τ such that $\text{rank}(\rho) \geq \beta$. By the universal property we know that there are ω many children $(\rho_i)_{i \in \mathbb{N}}$ of σ such that $\text{rank}(\rho_i) = \beta$. By Lemma 8.6 we have that $\beta = \text{rank}(\langle \rho_i, \rho \rangle)$, so there are ω many children of $\langle \sigma, \tau \rangle$ of rank β . $\dashv$

We now show that the universal tree as defined exists, and in fact has certain properties which would be useful in our proof of Theorem 8.1.

THEOREM 8.8. *There is a computable copy, $\mathcal{U}$, of the universal tree, such that this copy has a computable infinite path p . Moreover, this result relativizes: there is a computable functional $\mathcal{U}$ such that for each $X \subseteq \mathbb{N}$, we have that $\mathcal{U}^X$ has the universal tree property with respect to ω_1^X and that p is also a path in $\mathcal{U}^X$.*

For a proof of Theorem 8.8, we will use a relativised construction of a Harrison order, $\mathcal{H} \cong \omega_1^{CK} \cdot (1 + \eta)$. Thanks to Goh Jun Le, for pointing out that Harrison orders can be constructed in this uniform manner.

THEOREM 8.9 (Folklore). *There is a computable functional $\mathcal{H}$ such that $\mathcal{H}^X$ codes a linear order on $\mathbb{N}$ and $\mathcal{H}^X$ has no X -hyperarithmetic infinite descending sequence and has order type $\omega_1^X \cdot (1 + \eta)$.*

PROOF OF THEOREM 8.8. Fix a computable copy of the rationals $\mathbb{Q}$ and define T to be the tree of descending sequences in $\mathcal{H}^X \cdot (* + \mathbb{Q})$. Observe that $(0, 0), (0, -1), (0, -2), \dots$ is a computable infinite descending sequence in $\mathcal{H}^X \cdot (1 + \mathbb{Q})$, so gives us a computable path in T . We define U^X to be the tree $T \times \mathbb{N}^{<\mathbb{N}}$.

Clearly U^X is X -computable and has a computable infinite path, so it remains to check that it is universal relative to X . Observe that a node $\sigma \in T$ has ordinal rank if and only if the last bit of σ is of the form $(n, *)$ for some n . Using induction on $\alpha = \mathcal{H}^X \upharpoonright \{m <_{\mathcal{H}^X} n\}$ we get that $\text{rank}(\sigma) = \alpha$ in this case. This means that if $\text{rank} \sigma < \infty$, then $\text{rank} \sigma < \omega_1^X$ and for all $\beta < \min\{\text{rank} \sigma, \omega_1^X\}$ there is a child of σ of rank β . Since U^X is $T \times \mathbb{N}^{<\mathbb{N}}$, each node in U will have infinitely many such children.

Hence U^X has the universal property, as desired. $\dashv$

We now have all the tools we need to determine the degree of **cat**.

PROOF OF THEOREM 8.1. For this proof we will view $\mathbf{C}_{\mathbb{N}^{\mathbb{N}}}$ as the problem whose instances are ill-founded trees on $\mathbb{N}^{<\mathbb{N}}$ and whose solutions are paths in those trees. We start with the direction that uses the universal tree.

PROOF OF $\mathbf{C}_{\mathbb{N}^{\mathbb{N}}} \leq_W \mathbf{cat}$. Consider some ill founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$. We can view T as a tree in the language of directed graphs by defining $\sigma \frown i$ as the child of σ for $\sigma \frown i \in T$. Since this relation is computable relative to T , Lemma 8.7 tells us that $T \times \mathcal{U}^T \cong \mathcal{U}^T$. So we can use Theorem 8.8 to produce our instances of **cat**. Since U^T has a uniformly computable infinite path, any isomorphism will allow us to uniformly compute an infinite path in T . $\dashv$

PROOF OF $\mathbf{cat} \leq_W \mathbf{C}_{\mathbb{N}^{\mathbb{N}}}$. Consider some instance $p \oplus q$ of **cat**. Similar to what we did in the proof Theorem 2.2, we will consider a tree, T , partial isomorphism as a subset of $\mathbb{N}^{<\mathbb{N}}$. We will define T by saying that $f \oplus g \in [T]$ if

- $g = f^{-1}$. Which is $\forall n, m[f(n) = m \leftrightarrow g(m) = n]$.
- f is a monomorphism from p to q . That is $\forall e, \bar{a}[R_e^p(\bar{a}) \iff R_e^q(f(\bar{a}))]$

Since these are both Π_1 conditions relative to p and q , they will allow us to define a $p \oplus q$ -computable T with the desired properties. By definition, a path in T is an isomorphism, so we have the desired reduction. $\dashv$

So these two problems are equivalent. $\dashv$

§9. On top for cat. In computable structure theory, the study of universal classes of structures has been a big part of the field, and provides a tool for transferring results from one class of structures to another. One of the stronger way a class can be universal is via effective bi-interpretability [35]. For our purposes, the equivalent definition in terms of functors [31] is more useful:

DEFINITION 9.1. We say that class $\mathbb{C}$ is *uniformly bi-interpretable* in a class $\mathbb{K}$ if there is $\mathbb{K}' \subseteq \mathbb{K}$ and computable functors $F : \mathbb{C} \rightarrow \mathbb{K}'$ and $G : \mathbb{K}' \rightarrow \mathbb{C}$, such that F and G are *Pseudo inverses*.

A class, $\mathbb{K}$, of structures over a computable language, is *on top for effective bi-interpretability* if every other class of structures over a computable language is uniformly bi-interpretable in $\mathbb{K}$.

It is well known that graphs are on top for effective bi-interpretability, and we could have used this fact to provide definition of the general cat problem that used only one relation.

Another way a class can be universal is in terms of Borel or computable reductions:

DEFINITION 9.2 ([24], [9], [33]). A class of structures $\mathbb{K}$ is Borel (computably) reducible to a class $\mathbb{J}$, written $\mathbb{K} \leq_B \mathbb{J}$ ($\mathbb{K} \leq_c \mathbb{J}$), if there is a Borel (computable) function $f : \mathbb{K} \rightarrow \mathbb{J}$ that preserves equivalence under isomorphism, that is $\mathcal{M} \cong \mathcal{N} \iff f(\mathcal{M}) \cong f(\mathcal{N})$.

We say a class $\mathbb{K}$ is *on top for Borel reducibility* (*on top for computable reducibility*) if there is a Borel (computable) reduction, from the class of all countable structures to $\mathbb{K}$.

Due to the nature of Σ_1^1 -complete classes, it turns out that on top for Borel reducibility implies on top for computable reducibility.

While the reductions are on single structures, there is a notion of universal classes that considers pairs of structures, like we do considering $\text{cat}(\mathcal{M})$.

DEFINITION 9.3. We say that a set $X \subseteq \mathbb{N}^{\mathbb{N}}$ is Σ_1^1 -complete if for every Σ_1^1 -set $Y \subseteq \mathbb{N}^{\mathbb{N}}$ there is a computable function $\Phi : \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ such that $\Phi^{-1}[X] = Y$ and $\Phi^{-1}[X^c] = Y^c$.

For a class $\mathbb{K}$ we say that its isomorphism problem is Σ_1^1 -complete if the set $\{(A, B) \in \mathbb{K}^2 : A \cong B\}$ is Σ_1^1 -complete.

These universally notions are strictly ordered with $\mathbb{K}$ being on top for effective bi-interpretability implying it is on top for Borel reducibility, and $\mathbb{K}$ being on top for Borel reducibility implying that the isomorphism problem of $\mathbb{K}$ is Σ_1^1 -complete. These implications do not reverse, for instance trees are on top for Borel reducibility but not effective bi-interpretability, and torsion Abelian groups are not on top for Borel reducibility, but do have a Σ_1^1 -complete isomorphism problem.

In this section we consider a new universality notion, that of being on top for **cat**, and compare it to the above notions.

DEFINITION 9.4. Given a class of structures $\mathbb{K}$ we define $\text{cat}(\mathbb{K}) = \bigoplus_{\mathcal{M} \in \mathbb{K}} \text{cat}(\mathcal{M})$. We say that a class $\mathbb{K}$ is *on top for cat* if $\text{cat} \leq_W \text{cat}(\mathbb{K})$.

From the proof of Theorem 2.2 we immediately get that directed trees are on top for **cat**. Even undirected trees without labeled roots are on top for **cat** as in a computable tree, the unique path

between vertices a and b is computable uniformly in a, b . This immediately shows us that being on top for **cat** does not imply being on top for effective bi-interpretability.

In fact being on top for **cat** does not imply having a Σ_1^1 -complete isomorphism problem as the class of universal trees $\mathbb{U} = \{\mathcal{U}^X : X \subseteq \mathbb{N}\}$ is on top for **cat**, but only has one computable structure, so cannot have Σ_1^1 -complete isomorphism problem.

The following are some examples of classes that are on top for **cat**:

- Trees. Section 8.
- Relative Harrison orders, i.e. the class $\{\omega_1^X \cdot (1 + \eta) : X \subseteq \mathbb{N}\}$.
- Boolean algebras.
- Torsion Abelian groups. Theorem 9.5.
- Torsion free Abelian groups.

We give an explicit proof for the case of Torsion Abelian groups below. The other classes can be shown to be on top for **cat** using a similar approach, and adapting the proof of Σ_1^1 -complete isomorphism problem. For Harrison orders, we direct the reader to [36], for Boolean algebras to [10], and for torsion free Abelian groups to [38]. A careful reading of these papers should convince the reader that these classes are all on top for **cat**.

Some examples of classes that are not on top for **cat** are:

- Well orders.
- The rigid structures, $\{\mathcal{M} : |\text{aut}(\mathcal{M})| = 1\}$.
- Computable structures. Follows from Theorem 9.8, or alternatively Section 10.
- Equivalence relations. Follows from the uniformity of the proof of Theorem 6.15.

The classes of Well orders and rigid structures are both equivalent to $\text{UC}_{\mathbb{N}^{\mathbb{N}}}$, the restriction of $\text{C}_{\mathbb{N}^{\mathbb{N}}}$ to trees with exactly one infinite path. This follows from the fact that the partial isomorphism tree of a pair of rigid structures has at most one isomorphism and work done by Kihara, Marcone and Pauly [32].

THEOREM 9.5. *Torsion Abelian groups are on top for cat.*

PROOF. We start by defining how to interpret a tree on $\mathbb{N}^{<\mathbb{N}}$ as a torsion group. We start with the product of the torsion group of dyadic rationals modulo 1 and define a subgroup by listing a generating set.

Fix a tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$. Define the function $f : T \rightarrow \mathbb{Q}^{\mathbb{N}}$ as follows:

$$f(\sigma)(\lceil \tau \rceil) = \begin{cases} 2^{-|\sigma|+|\tau|+1}, & \tau \preceq \sigma, \\ 0, & \text{otherwise.} \end{cases}$$

We define our torsion group G_T to be the closure of the range of f . Observe that $f(\sigma)(n) = 0$ for all but finitely many n . This allows us to require our presentation of our torsion group G_T , have the length of each element be computable. Another important observation is that $f(\sigma \cap n) +_G f(\sigma \cap n) = f(\sigma) \neq f(\sigma \cap n)$ and $f(\emptyset) + f(\emptyset) = 0$. This means that any element of G_T can be uniquely written as $\sum_{\sigma \in F} f(\sigma)$ for some finite set $F \subseteq T$.

Now consider an isomorphism $h : G_T \rightarrow G_S$ for some isomorphic trees T and S . We will now show that uniformly in T, S, h and an infinite path $p \in [T]$ we can compute an infinite path in S .

Consider the computable sequence of sets $\{F_n\}_{n \in \mathbb{N}}$ defined by $\sum_{\sigma \in F_n} f_S(\sigma) = h(f_T(p \upharpoonright n))$. Since $h(f_T(p \upharpoonright n)) + h(f_T(p \upharpoonright n)) = h(f_T(p \upharpoonright (n-1)))$, it must be that $F_{n-1} \subseteq \{\tau \prec \sigma : \sigma \in F_n\}$. So if $\sigma \in F_n$ then there must be an extension of σ in F_{n+1} . This allows us to compute an infinite path in S . $\dashv$

We have claimed that if $\mathbb{K}$ is on top for effective bi-interpretability, then $\mathbb{K}$ is on top for $\mathbf{cat}$, and while intuitive, this does require a proof. We will prove a stronger result that allows us to use effective bi-interpretations as a tool for proving that a class is on top for $\mathbf{cat}$.

THEOREM 9.6. *If $\mathbb{C}$ is effectively bi-interpretable in $\mathbb{K}$, then $\mathbf{cat}(\mathbb{C}) \leq_W \mathbf{cat}(\mathbb{K})$.*

PROOF. Let (Φ, Φ_*) be a computable representation of the functor $F : \mathbb{C} \rightarrow \mathbb{K}'$, (Ψ, Ψ_*) be a computable representation of the functor $G : \mathbb{K}' \rightarrow \mathbb{C}$, and Λ is a computable functional showing that $G \circ F$ is effectively isomorphic to the $\text{id}_{\mathbb{C}}$. Define $\Phi_2 : \mathbb{C}^2 \rightarrow \mathbb{K}^2$ by $\Phi_2(\mathcal{M}_0, \mathcal{M}_1) = (\Phi(\mathcal{M}_0), \Phi(\mathcal{M}_1))$. Φ_2 will be the forward functional. Define $\Psi_b : \mathbb{C}^2 \times \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ by $\Psi_b(\mathcal{M}_0, \mathcal{M}_1, f) = \Psi_*(\Phi(\mathcal{M}_0), \Phi(\mathcal{M}_1), f)$. Our backward functional will be $\Lambda \circ \Psi_b$.

The rest of the proof is just definition chasing. $\dashv$

COROLLARY 9.7. *Let $\mathbb{U}$ be the class of structures $\{\mathcal{U}^X : X \subseteq \mathbb{N}\}$. If $\mathbb{U}$ is effectively bi-interpretable in $\mathbb{K}$, then $\mathbb{K}$ is on top for $\mathbf{cat}$.*

We do not know if the above corollary reverses.

9.1. Relationship to Σ_1^1 -complete classes. The reason that a class $\mathbb{K}$ can be on top for $\mathbf{cat}$, but without Σ_1^1 -complete isomorphism problem comes down to the fact that the reduction that shows $\mathbb{K}$ is on top for $\mathbf{cat}$ only needs to produce isomorphic pairs, where as a reduction from a Σ_1^1 complete set, $A \subseteq \mathbb{N}^{\mathbb{N}}$, to $\mathbb{K}^2$ needs to always produce members of $\mathbb{K}^2$, even on inputs $x \notin A$. We now show that this is essentially the only reason for a lack of an implication.

THEOREM 9.8. *If $\mathbb{C}_{\mathbb{N}^{\mathbb{N}}} \leq_W \mathbf{cat}(\mathbb{K})$ via forward functional Φ and $\text{range}(\Phi) \subseteq \mathbb{K}^2$, then the isomorphism problem for $\mathbb{K}$ is Σ_1^1 -complete.*

PROOF. We will use the fact that the collection of ill-founded trees is a Σ_1^1 -complete class.

Consider $\text{dom}(\Phi)$. As a forward functional we know this includes all ill-founded trees, but there is no requirement that it contains every tree. However, since Φ is computable, we know that its domain is a Π_2^0 class.

Let Ψ be the backward functional used by the reduction $\mathbb{C}_{\mathbb{N}^{\mathbb{N}}} \leq_W \mathbf{cat}(\mathbb{K})$. Consider the set:

$$\{T : T \notin \text{dom}(\Phi) \vee \exists f[\Phi(T)_0 \simeq_f \Phi(T)_1 \wedge \Psi(f) \notin \llbracket T \rrbracket]\}.$$

This set is by definition Σ_1^1 and contains only T on which the reduction using Φ and Ψ does not work, i.e. only well-founded trees. By Σ_1^1 -bounding there is some computable ordinal $\alpha < \omega_1^{CK}$ such that all trees in this set have rank $< \alpha$.

Fix a computable tree S of rank α . Define a computable function $\Theta : \text{trees} \rightarrow \mathbb{K}^2$ by $\Theta(T) = \Phi(T \sqcup S)$. By assumption, $\text{dom}(\Theta) = \text{trees}$ and if T is ill-founded, then $\Theta(T)$ is a pair of isomorphic structures in $\mathbb{K}^2$. If $(A, B) = \Theta(T)$ and $f : A \rightarrow B$ is an isomorphism then it must be that $\Psi(f)$ is a path in $T \sqcup S$ and so T is ill-founded. Hence Θ is a reduction from ill-founded trees to $\{(A, B) \in \mathbb{K}^2 : A \cong B\}$. $\dashv$

QUESTION 9.9. Is there a class $\mathbb{K}$ that has Σ_1^1 -complete isomorphism problem, but is not on top for $\mathbf{cat}$?

Towards answering this question we have a candidate class that is Σ_1^1 -complete and resists our previously used techniques of showing that a class is on top for $\mathbf{cat}$. We describe this class now. The idea is to take the class of universal trees, i.e. those in which every possible rank occurs, and ensure that any infinite path in an ill-founded tree in the class must compute an infinite descending sequence in a Harrison order. In this way the class will be Σ_1^1 -complete for the same reasons that

trees are, but it will lack nice trees with computable infinite paths, so from a generic isomorphism it will be hard to compute a infinite path.

Fix a computable Harrison order $\mathcal{H} = (\mathbb{N}, <_{\mathcal{H}})$, that is, a computable linear order with no hyperarithmetic infinite descending sequence. We define our class, $\mathbb{K}$, as a subset of the class of directed graphs with labeled edges. For each structure $\mathcal{T} \in \mathbb{K}$ we require:

- For each $x \in T$, there is at most one $n \in \mathbb{N}$ and $w \in T$ such that $E_n(w, x)$ holds.
- There is a unique root, r , such that for all x and n , $\neg E_n(x, r)$.
- For each n , E_n is antisymmetric and irreflexive.
- If $E_n(x, y) \wedge E_m(y, z)$ then $m <_{\mathcal{H}} n$.

We define the rank of a point $x \in T$ using the usual recursive definition: if for all y and n , $\neg E_n(x, y)$, then $\text{rank}(x) = 0$, otherwise $\text{rank}(x) = \sup \text{rank}(y) + 1 : \exists n[E_n(x, y)]$, and if x can be extended to an infinite path then $\text{rank}(x) = \infty$.

We require that all possible ranks occur as much as possible: for each $w, x \in T$, $n \in \mathbb{N}$ with $E_n(w, x)$ we require that for $\beta < \text{rank}(x)$ and $m <_{\mathcal{H}} n$ such that $\text{ot}(\{\ell : \ell <_{\mathcal{H}} m\}) \geq \beta$ there are infinitely many $y \in T$ such that $E_m(x, y)$ and $\text{rank}(y) = \beta$. For the root, r , we simply require that for all $\beta < \text{rank}(r)$ and m such that $\text{ot}(\{\ell : \ell <_{\mathcal{H}} m\}) \geq \beta$ there are infinitely many $y \in T$ such that $E_m(r, y)$ and $\text{rank}(y) = \beta$.

To make this a reasonable definition we restrict ranks to be computable ordinals relative to some oracle X .

To see that the isomorphism problem for this class is Σ_1^1 -complete, one can observe that if we take a tree S , we can build a tree in $\mathbb{K}$ by defining domain to be the tree $(S \times \mathcal{U}^S) \times (DS(\mathcal{H}) \times \mathbb{N}^{<\mathbb{N}})$ and define the edge relation by $E_n(\sigma, \sigma \frown a)$ if the $\mathcal{H}$ part of a is n . This tree in $\mathbb{K}$ is ill-founded if and only if S is, in which case it will be isomorphic to the tree $\mathcal{U}^S \times (DS(\mathcal{H}) \times \mathbb{N}^{<\mathbb{N}})$.

§10. An upper bound on $\text{cat}(M)$ for computable M . One motivating reason to study the general cat problem from Section 8 is that it is clearly an upper bound on $\text{cat}(M)$ for any M . In this section, we give a tight upper bound for $\text{cat}(M)$ when M is computable, by showing that cat_0 has a maximal elements. We also show that there is no maximal $\text{cat}(M)$ in general.

The natural candidate for an upper bound is of course the universal tree $\mathcal{U}$ from Section 8.

THEOREM 10.1. *If M is a computable structure then $\text{cat}(M) \leq_{\text{w}} \text{cat}(\mathcal{U})$.*

PROOF. Constructing the reduction is not hard. Given two copies p and q of M we construct the tree of partial isomorphisms, T , from p to q and use $T \times \mathcal{U}, \mathcal{U}$ as our instance of $\text{cat}(\mathcal{U})$. As explained in the proof of Theorem 8.1, provided $T \times \mathcal{U} \cong \mathcal{U}$, an isomorphism between these two trees will allow us to compute a path in the tree of partial isomorphisms T , and thus an isomorphism from p to q .

The tricky part comes in verifying that $T \times \mathcal{U}$ is actually isomorphic to $\mathcal{U}$. If p and q are computable then so is T , and so this holds by Lemma 8.7. However, even though M is computable, there is no requirement that p and q , and thus T , are also computable. So we now show that T has no nodes of rank between ω_1^{CK} and ∞ . If we can do this then Lemma 8.7 will apply.

Fix isomorphisms f and g from M (a computable copy), to p and q respectively. Let S be the standard tree of partial automorphisms of M , and T be the tree of partial isomorphisms from p to q . Given some node $\sigma = \sigma_0 \oplus \sigma_1 \in T$ we define $\text{Fix}(S, \sigma)$ to be the subtree of S consisting of nodes $\tau = \tau_0 \oplus \tau_1$, such that τ_0 is compatible with $f \circ \sigma_0 \circ g^{-1}$ and τ_1 is compatible with $g \circ \sigma_1 \circ f^{-1}$, i.e. $\text{Fix}(S, \sigma)$ consist of all partial isomorphisms that do not disagree with σ .

CLAIM 10.1. *For all $\sigma \in T$, if $\text{rank}(\sigma) < \infty$ then $\text{rank}(\sigma) < \text{rank}(\text{Fix}(S, \sigma)) + \omega$.*

PROOF. We will use induction on the rank of σ . The base case is trivial: If σ is a leaf then $\text{rank}(\sigma) = 0 < \omega \leq \text{rank}(\text{Fix}(S, \sigma)) + \omega$.

Now suppose the claim holds for all nodes of rank less than α for some ordinal α and let $\sigma \in T$ be such that $\text{rank}(\sigma) = \alpha$. Then for each $\sigma \frown i \in T$ we have that

$$\text{rank}(\sigma \frown i) < \text{rank}(\text{Fix}(S, \sigma \frown i)) + \omega \leq \text{rank}(\text{Fix}(S, \sigma)) + \omega.$$

If α is a successor ordinal then there is some i such that $\alpha = \text{rank}(\sigma \frown i) + 1$, so the claim holds.

Suppose that α is a limit ordinal. Then we have that $\alpha \leq \text{rank}(\text{Fix}(S, \sigma)) + \omega$. Suppose towards a contradiction that $\alpha = \text{rank}(\text{Fix}(S, \sigma)) + \omega$. Let γ be a limit ordinal and $k \in \omega$ be such that $\text{rank}(\text{Fix}(S, \sigma)) = \gamma + k$. Consider some node $\tau \in \text{Fix}(S, \sigma)$ with $|\tau| > k$. By definition of rank it must be that $\text{rank}(\tau) < \gamma$.

Now let m be large enough that $0, \dots, k \in f(\{0, \dots, m\}) \cap g(\{0, \dots, m\})$. Consider some extension $\rho \succ \sigma$ with $|\rho| > 2m$. Since $\rho = \rho_0 \oplus \rho_1$ is sufficiently long it must be that $0, \dots, k \in \text{dom}(f \circ \rho_0 \circ g^{-1}) \cap \text{dom}(g \circ \rho_1 \circ f^{-1})$. So when we consider the tree $\text{Fix}(S, \rho)$ it must not branch until we reach nodes of length $\geq k$. Since $\text{Fix}(S, \rho) \subseteq \text{Fix}(S, \sigma)$, we must have that $\text{rank}(\text{Fix}(S, \rho)) < \gamma$, so by induction we have that $\text{rank}(\rho) < \text{rank}(\text{Fix}(S, \rho)) + \omega \leq \gamma$.

Since ρ was an extension of length $> 2m$ this means that $\text{rank}(\sigma) \leq \gamma$, but that is a contradiction. So we have that $\text{rank}(\sigma) < \text{rank}(\text{Fix}(S, \sigma)) + \omega$. $\dashv$

So we have shown that $\text{rank}(\sigma)$ is always bounded by a computable ordinal for any $\sigma \in T$ of rank $< \infty$. Thus we have that $T \times \mathcal{U} \cong \mathcal{U}$, and the reduction is valid. $\dashv$

So we have shown that $\text{cat}(\mathcal{U})$ is universal for computable structures. What about non computable structures? The following result show that there is no way to generalize this.

THEOREM 10.2. $\text{cat}(\mathcal{U}^\mathcal{O}) \not\leq_W \text{cat}(\mathcal{U})$.

PROOF. Suppose there is a reduction given by the functionals Φ and Ψ . It is known that there is an ill-founded, $\mathcal{O}$ -computable tree T with no $\Delta_1^1(\mathcal{O})$ infinite path (e.g. the tree of infinite descending sequences in a relativised Harrison order). The instance we will use to get a contradiction will be $\mathcal{U}^\mathcal{O}, T \times \mathcal{U}^\mathcal{O}$.

Let $(p, q) = \Phi(\mathcal{U}^\mathcal{O}, T \times \mathcal{U}^\mathcal{O})$. Consider the tree S of partial isomorphism from p to q , and consider its left most path, f . Since S is $\mathcal{O}$ -computable, we have that f is $\Pi_1^1(\mathcal{O})$. By Claim 10.1 we have that all nodes in S are of rank ∞ or $< \omega_1^{CK}$. However this means that f is $\Sigma_1^1(\mathcal{O})$ (for all $\sigma <_{\text{lex}} f$ there is an embedding of the tree above σ into a well-founded computable tree).

Since f is an isomorphism from p to q , it must be that $\Psi(f, (\mathcal{U}^\mathcal{O}, T \times \mathcal{U}^\mathcal{O}))$ is an isomorphism from $\mathcal{U}^\mathcal{O}$ to $T \times \mathcal{U}^\mathcal{O}$. However $\Psi(f, (\mathcal{U}^\mathcal{O}, T \times \mathcal{U}^\mathcal{O}))$ can $\mathcal{O}$ -compute an infinite path in T and $\Psi(f, (\mathcal{U}^\mathcal{O}, T \times \mathcal{U}^\mathcal{O}))$ is $\Delta_1^1(\mathcal{O})$, so we have a contradiction. $\dashv$

COROLLARY 10.3. *There is no model $\mathcal{M}$ such that for all models $\mathcal{N}$ we have $\text{cat}(\mathcal{N}) \leq_W \text{cat}(\mathcal{M})$.*

PROOF. If there was such a model $\mathcal{M}$ then it would have to be that $\text{id} \leq_W \text{cat}(\mathcal{M})$. So $\text{cat}(\mathcal{M})$ would have to have computable instances and hence $\mathcal{M}$ would be computable. So $\text{cat}(\mathcal{M}) \leq_W \text{cat}(\mathcal{U})$, a contradiction of Theorem 10.2. $\dashv$

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SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES
 NANYANG TECHNOLOGICAL UNIVERSITY
 21 NANYANG LINK, SINGAPORE
E-mail: nickolay.bazhenov@ntu.edu.sg

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES
 NANYANG TECHNOLOGICAL UNIVERSITY
 21 NANYANG LINK, SINGAPORE
E-mail: josiah.jg@ntu.edu.sg

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES
 NANYANG TECHNOLOGICAL UNIVERSITY
 21 NANYANG LINK, SINGAPORE
E-mail: liling.ko@ntu.edu.sg

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
 UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA
 NO. 2006 XIYUAN AVE, WEST HI-TECH ZONE, CHENGDU, P.R. CHINA
E-mail: heertern001@e.ntu.edu.sg

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES
 NANYANG TECHNOLOGICAL UNIVERSITY
 21 NANYANG LINK, SINGAPORE
E-mail: kmng@ntu.edu.sg